Thesis

Algorithms and Iterative Methods for Infinite Games

Tomáš Votroubek

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Czech Technical University in Prague

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Contents

1. Applications in Continuous Games

Continuous games are multiplayer games in which strategy sets are compact and utility functions are continuous. Many application domains have a continuum of actions, such as the allocation of time, resources, location in space [30], or parameters of classifiers [68]. This also includes several games modeling cybersecurity scenarios [46, 68, 56].

Continuous games have equilibria in mixed strategies by Glicksberg's generalization of Nash's theorem [22], but these equilibria are typically very hard to characterize and compute. We point out the main difficulties in the development of algorithms and numerical methods for continuous games:

- The equilibrium can be any tuple of mixed strategies with infinite supports [32] or almost any tuple of finitely-supported mixed strategies [54].
- Bounds on the size of supports of equilibrium strategies are known only for particular classes of continuous games [58].
- Some important games have only mixed equilibria; for example, certain variants of Colonel Blotto games [23].
- Finding a mixed strategy equilibrium involves locating its support, which lies inside a continuum of points.

To the best of our knowledge, algorithms and numerical methods exist only for very special classes of continuous games. In particular, two-player zero-sum polynomial games can be solved using sum-of-squares optimization based on a sequence of semidefinite relaxations of the original problem [49, 35]. Başar and Olsder [5] have written a detailed analysis of equilibria for some families of games with a particular shape of utility functions (e.g., games of timing, bell-shaped kernels). Fictitious play, one of the principal learning methods for finite games, was recently extended to continuous action spaces and applied to Blotto games [21]. The dynamics of fictitious play have been analyzed only under further restrictive assumptions in the continuous setting [28]. No-regret learning, studied by Mertikopoulos and Zhou [40], can be applied to finding pure equilibria or to mixed strategy learning in finite games. Convergence guarantees for algorithms in the distributed environment solving convex-concave games and some generalizations thereof have been developed by Mertikopoulos et al. [41]. In a similar setting, Chasnov et al. [11] provides convergence guarantees to a neighborhood of a stable Nash equilibrium for gradient-based learning algorithms.

1.1. Continuous Games and Nash Equilibria

The player set is denoted by $N = \{1, \dots, n\}$. Each player $i \in N$ selects a pure strategy x_i from a nonempty compact set $X_i \subseteq \mathbb{R}^{d_i}$, where d_i is a positive integer. A pure strategy profile is the n-tuple $\mathbf{x} = (x_1, \dots, x_n) \in \mathbf{X}$, where $\mathbf{X} = X_1 \times \cdots \times X_n$. We assume that each utility function $u_i : \mathbf{X} \to \mathbb{R}$ is continuous. A *continuous game* is the tuple

$$\mathcal{G} = \langle N, (X_i)_{i \in N}, (u_i)_{i \in N} \rangle.$$

A mixed strategy of player i can be an arbitrary Borel probability measure μ_i over X_i. Note that this general framework allows us to consider mixed strategies whose support is uncountably infinite, which is inevitable for the coherent treatment of games over compact strategy sets.

By M_i we denote the space of all mixed strategies of player $i \in N$. Let $\mathbf{M} = M_1 \times \cdots \times M_n$ and let $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n) \in \mathbf{M}$ be a profile of mixed strategies. For the sake of brevity, we use $\boldsymbol{\mu}$ in place of the corresponding product probability measure $\mu_1 \times \cdots \times \mu_n$ in the definitions below involving integrals. The expected utility of player $i \in N$ is the function $U_i : \mathbf{M} \to \mathbb{R}$

$$U_i(\boldsymbol{\mu}) = \int_{\mathbf{X}} u_i \, \mathrm{d}\boldsymbol{\mu}$$

This definition yields a function $U_i : \mathbf{M} \to \mathbb{R}$, which can be effectively evaluated only in special cases, such as when the support of each μ_i is finite.

We define the sets which exclude one player as

$$\mathbf{X}_{-i} = \bigtimes_{\substack{j \in N \\ j \neq i}} X_j \quad \text{and} \quad \mathbf{M}_{-i} = \bigtimes_{\substack{j \in N \\ j \neq i}} M_j.$$

For every $\boldsymbol{\mu} \in \mathbf{M}$ and each player $i \in N$, by $\boldsymbol{\mu}_{-i}$ we denote the restriction of $\boldsymbol{\mu}$ onto $\mathbf{M}_{-i}$. The value of expected utility of player i in a strategy profile $(x_i, \boldsymbol{\mu}_{-i}) \in X_i \times \mathbf{M}_{-i}$ is denoted by

$$U_i(x_i, \boldsymbol{\mu}_{-i}) = \int_{\mathbf{X}_{-i}} u_i(x_i, .) \, \mathrm{d}\boldsymbol{\mu}_{-i}. \tag{1.1}$$

Definition 1 (Nash [43])**.** *A Nash equilibrium is a strategy profile $\boldsymbol{\mu}^\star \in \mathbf{M}$ satisfying*

$$U_i(\boldsymbol{\mu}^\star) = \max_{x_i \in X_i} U_i(x_i, \boldsymbol{\mu}^\star_{-i}), \qquad \forall i \in N.$$

That is, each player's strategy is a best response against those of the others.

Theorem 1 (Glicksberg [22])**.** *Every continuous game has a mixed strategy Nash equilibrium.*

1.2. Nash Equilibria in Separable Network Games

Separable network games [8, 10] are multiplayer strategic games in which players interact only with adjacent players in a simple undirected graph. The utility of each player results from the aggregation of utilities in the corresponding general-sum two-player games. We extend this model to infinite *network games* whose strategy sets are compact subsets of the Euclidean space. We show that *Nash Equilibria* (NE) of a zero-sum continuous network game can be characterized as optimal solutions to a specific infinite-dimensional linear optimization problem. In particular, when the utility functions are multivariate polynomials, this optimization formulation enables us to approximate the equilibria using a hierarchy of semidefinite relaxations based on the Moment-SOS hierarchies discussed in **??**.

1.2.1. Continuous Network Games

Separable network games can capture situations in which players distribute their capacities or resources in a continuous way across multiple targets to achieve control over them. The targets can be battlefields, inspection sites, exit points etc.

In the class of games considered in this chapter, each player $i \in N$ is identified with a node of an undirected graph (N, E) without loops. The presence of an edge $\{i, j\} \in E$ means that players i and j engage in a two-person general-sum strategic game in which the utility functions of players i and j are functions $u_{ij} : X_i \times X_j \to \mathbb{R}$ and $u_{ji} : X_j \times X_i \to \mathbb{R}$.

We will always assume that the functions u_{ij} and u_{ji} are at least integrable, ensuring that every expected utility appearing below is correctly defined. Typically, a player i participates in multiple pairwise games, in which case, the same strategy strategy $x_i \in X_i$ is implemented in all two-person games played with the adjacent players in the graph (N, E). Thus, the aggregated utility of player i is computed as

$$u_i(\mathbf{x}) = \sum_{\substack{j \in N \\ \{i,j\} \in E}} u_{ij}(x_i, x_j), \qquad \mathbf{x} \in \mathbf{X}. \tag{1.2}$$

The resulting n-person game

$$\mathcal{G} = \langle N, (X_i)_{i \in N}, (u_i)_{i \in N} \rangle \tag{1.3}$$

is called a *separable network game with compact strategy sets*. Throughout the paper we refer to game (1.3) as a “network game” for short. We say that $\mathcal{G}$ is

- a *polymatrix game* when all strategy sets X_i are finite;
- *zero-sum* if $\sum_{i \in N} u_i(\mathbf{x}) = 0$ for all $\mathbf{x} \in \mathbf{X}$;
- *continuous* if each utility function u_i is continuous.

The class of network games most studied in the literature is the class of zero-sum polymatrix games. The “global” zero-sum assumption and finite strategy sets make such games computationally tractable [10, 18, 4, 57, 9].

Example 1. *Let $N = \{1, 2, 3, 4\}$ and the graph (N, E) be the cycle on Figure 1.1. This is the smallest example of a network game such that each player interacts only with a strict subset of other players. For example, the utility function of player 3 is $u_3(x_1, x_2, x_3, x_4) = u_{31}(x_3, x_1) + u_{34}(x_3, x_4)$. The zero-sum condition for this game can be formulated as $u_3 = -u_1 - u_2 - u_4$.*

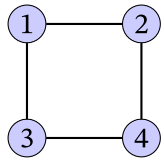


Figure 1.1.: A network game with 4 players and 4 pairwise games

As a direct consequence of Glicksberg's theorem, every continuous network game $\mathcal{G}$ has a NE $\boldsymbol{\mu}^\star \in \mathbf{M}$ in mixed strategies. We will show that the NE in a zero-sum continuous network game can be characterized using an optimization-based approach. Specifically, we consider the following infinite-dimensional linear optimization problem, which generalizes *Linear Program 1* of Cai et al. [10]:

$$\begin{aligned}
\underset{\boldsymbol{\mu}, \mathbf{w}}{\text{minimize}} \quad & \sum_{i \in N} w_i && (1.4a)\\
\text{subject to} \quad & w_i \geq U_i(x_i, \boldsymbol{\mu}_{-i}) \qquad \forall i \in N \;\; \forall x_i \in X_i, && (1.4b)\\
& \boldsymbol{\mu} = (\mu_1, \dots, \mu_n) \in \mathbf{M}, && (1.4c)\\
& \mathbf{w} = (w_1, \dots, w_n) \in \mathbb{R}^n && (1.4d)
\end{aligned}$$

Theorem 2. *For every profile of mixed strategies $\boldsymbol{\mu}^\star \in \mathcal{P}$ in a zero-sum continuous network game $\mathcal{G}$, the following are equivalent:*

1. *$\boldsymbol{\mu}^\star$ is a Nash equilibrium of game $\mathcal{G}$.*
2. *$(\boldsymbol{\mu}^\star, \mathbf{w}^\star)$ is an optimal solution to (1.4), where*

$$w_i^\star = \max_{x_i \in S_i} U_i(x_i, \boldsymbol{\mu}_{-i}^\star), \qquad i \in N. \tag{1.5}$$

Proof. Firstly, observe that the expression $\max_{\nu_i \in P_i} U_i(\nu_i, \boldsymbol{\mu}_{-i})$ is maximized by a ν_i supported on any of U_i's maximizers. Therefore, given a feasible $(\boldsymbol{\mu}, \mathbf{w})$, the constraint (1.4b) implies

$$w_i \geq \max_{x_i \in X_i} U_i(x_i, \boldsymbol{\mu}_{-i}) = \max_{\nu_i \in P_i} U_i(\nu_i, \boldsymbol{\mu}_{-i}) \geq U_i(\mu_i, \boldsymbol{\mu}_{-i}) \quad \forall i \in N. \tag{1.6}$$

Note that the maxima in inequalities (1.6) exist by compactness of X_i and the continuity of the functions $U_i(\cdot, \boldsymbol{\mu}_{-i}) : X_i \to \mathbb{R}$. Applying inequalities (1.6) in the objective, together with the zero-sum property, gives:

$$\sum_{i \in N} w_i \geq \sum_{i \in N} U_i(\mu_i, \boldsymbol{\mu}_{-i}) = 0. \tag{1.7}$$

By Theorem 1, there exists an NE $\boldsymbol{\mu}^\star$ such that $U_i(\boldsymbol{\mu}^\star) = \max_{x_i \in X_i} U_i(x_i, \boldsymbol{\mu}_{-i}^\star)$ $\forall i \in N$. Let $\boldsymbol{\mu}$ be an NE and $\mathbf{w}_i^\star$ be as in condition (1.5), then the inequalities in (1.6) become equalities, and the lower bound of zero is achieved in (1.7).

We have shown that NE correspond to optimal solutions. Conversely, consider any optimal solution $(\boldsymbol{\mu}^\star, \mathbf{w}^\star)$. Notice that $w_i^\star$ must be tight upper bounds on $U_i(\cdot, \boldsymbol{\mu}_{-i}^\star)$. By the zero-sum property, $\sum_{i\in N} U_i(\boldsymbol{\mu}) = 0$ for any $\boldsymbol{\mu}$. If $\boldsymbol{\mu}^\star$ were not an NE, then there exists a player i such that $\max_{x_i\in X_i} U_i(x_i, \boldsymbol{\mu}_{-i}) > U_i(\boldsymbol{\mu})$. This implies that

$$0 = \sum_{i\in N} U_i(\boldsymbol{\mu}) < \sum_{i\in N} \max_{x_i\in X_i} U_i(x_i, \boldsymbol{\mu}_{-i}) = \sum_{i\in N} w_i,$$

i.e., the objective value is not optimal. □

The Borel probability measures and continuous functions in Problem 1.4 do not generally allow for effective computation. Still, there are cases where the solution can be computed. For example, in the case of zero-sum polymatrix games, Problem 1.4 becomes a linear program [10]. Similarly, if we restrict the model to polynomial functions and semialgebraic sets, the solution can be computed using the tools of polynomial optimization discussed in **??**.

1.2.2. Polynomial Network Games

In this section, we make the following assumptions about the strategy spaces and the utility functions of a network game $\mathcal{G}$ defined by (1.3).

1. Every compact strategy space $X_i \subseteq \mathbb{R}^{m_i}$ is a *basic semi-algebraic set*; that is,

$$X_i = \{\mathbf{x} \in \mathbb{R}^{m_i} \mid g_{ik}(\mathbf{x}) \geq 0,\ k \in K_i\}, \tag{1.8}$$

   where each g_{ik} is a real polynomial in m_i variables and $K_i \neq \emptyset$ is a finite index set. We also assume the standard archimedean certificate on compactness on the sets; e.g., that one of the constraints has the form $g_{ik}(\mathbf{x}) = a - \|\mathbf{x}\|^2$ for some $k \in K_i$ and some positive a.

2. In every two-player game involving players $i, j \in N$, the utility functions u_{ij} and u_{ji} of player i and j, respectively, are real polynomials in $m_i + m_j$ indeterminates. Therefore, each utility function u_i given by (1.2) is a real polynomial in $m_1 + \cdots + m_n$ indeterminates.

3. The game $\mathcal{G}$ is zero-sum.

These three assumptions define a *polynomial zero-sum network game* $\mathcal{G}$. We remark that the first assumption is very mild — many sets important for applications are in fact the solution sets of finitely many polynomial inequalities. In particular, a compact semi-algebraic set is not necessarily convex or connected. Moreover, Glicksberg's theorem (1) can be significantly strengthened for polynomial games — each polynomial game has NE in which every player mixes only among finitely many pure strategies [64, 58].

The solutions for polynomial zero-sum network games can be solved using Lasserre's hierarchies as described in **??**. Specifically, we relax the constraints on measures and polynomial inequalities of Program 1.4. Instead of requiring that the mixed strategies of players lie in the cone of probability measures, we apply constraints on truncated sequences of moments. Similarly, we replace the inequalities (1.4b) with the constraint that $w_i - U_i(\cdot, \boldsymbol{\mu}_{-i})$ lies in the truncated quadratic module.

The h-th level of the hierarchy is the following semidefinite program:

$$\begin{array}{lll} \underset{\hat{\mathbf{y}}_1,\dots,\hat{\mathbf{y}}_n,\mathbf{w}}{\text{minimize}} & w_1 + \cdots + w_n & \\ \text{subject to} & w_i - \bar{u}_i(\hat{\mathbf{y}}_{-i}) \in Q_{2h}(X_i) & \forall i \in N \\ & \hat{\mathbf{y}}_i = (y_{i,\alpha})_{\alpha \in \mathbb{N}^{m_i}_{2h}} \text{ and } y_{i,\mathbf{0}} = 1 & \forall i \in N \\ & M_h(\hat{\mathbf{y}}_i) \succeq 0 & \forall i \in N \\ & M_h(g_{ik}, \hat{\mathbf{y}}_i) \succeq 0 & \forall k \in K_i, \forall i \in N \\ & \mathbf{w} = (w_1, \dots, w_n) \in \mathbb{R}^n & \end{array} \tag{1.9}$$

We then iteratively solve each level of the hierarchy until all sequences $\hat{\mathbf{y}}_i$ have a representing atomic measure, which we extract using the algorithm of Lasserre [37, Algorithm 6.9]. With the strategies of other players fixed, the best response condition is equivalent to certifying optimality for a polynomial over a semialgebraic set.

1.2.3. Experiments with Polynomial Network Games

In the remaining part of this paper we show several examples of polynomial games and recovered their equilibria by the method explained in the Section 1.2.2. We implemented the hierarchy of programs (1.9) in Julia using JuMP [17] and the SumOfSquares.jl package [66]. We solved the semidefinite programs using Mosek on a laptop with a 3.1GHz dual-core CPU. Our code is available online.

Example (Example 39)**.** *Consider a two-player zero-sum polynomial game played on the $[-1,1]$ interval. The utility functions are*

$$u_1(x_1, x_2) = 2x_1x_2^2 - x_1^2 - x_2 = -u_2(x_1, x_2).$$

Since polynomial network games are a generalization of polynomial games, we can use the same framework to solve them. A pure NE is achieved when Player 1 plays $x_1 \approx 0.4$ and Player 2 plays $x_2 \approx 0.63$. The optimal payoffs to each player are $\mathbf{w} \approx (-0.47, 0.47)$. This problem was solved at order 4 of the hierarchy. Due to the low number of variables the problem required only 44 constraints and it was solved within 10 ms.

Parrilo [50] demonstrated that the hierarchy is not necessary for two-player zero-sum games with interval strategy sets. In this case, Lukács' theorem [63, p. 1.21.1] states that there exists a Putinar decomposition in terms of rank-1 multipliers such that the term degrees do not exceed that of the input polynomial. The $z \in [-1,1]$ strategy set would be encoded as $1 + z \geq 0 \wedge 1 - z \geq 0$ for inputs with odd degrees and $1 - z^2 \geq 0$ for even degrees. Both encodings satisfy the Archimedean property as the product of the defining polynomials is positive on a compact set. However, semidefinite solvers usually cannot guarantee rank-1 solutions.

Example (Example 48)**.** *We consider a three-player polynomial network game with a complete graph, where each strategy set is the interval $[-1,1]$ and the utility functions are*

$$\begin{aligned} u_1(x_1, x_2, x_3) &= -2x_1x_2^2 - 2x_1^2 + 5x_1x_2 - 4x_1x_3 - x_2 - 2x_3, \\ u_2(x_1, x_2, x_3) &= 2x_1x_2^2 - 2x_2x_3^2 - 2x_1^2 - 5x_1x_2 - 2x_2^2 + 5x_2x_3 + x_2, \\ u_3(x_1, x_2, x_3) &= 2x_2x_3^2 + 4x_1^2 + 4x_1x_3 + 2x_2^2 - 5x_2x_3 + 2x_3. \end{aligned}$$

The game has the following partially-mixed Nash equilibrium:

$$\mu_1^\star \approx \delta_{-0.06},$$
$$\mu_2^\star \approx \delta_{0.35},$$
$$\mu_3^\star \approx 0.72\delta_1 + 0.28\delta_{-1}.$$

The corresponding utilities are $\mathbf{w} \approx (-1.22, 0.26, 0.97)$*. This problem converged in around* 10 *ms at order 4, resulting in 96 constraints.*

Example (Example 49)**.** *We consider a polynomial security game with attackers* $N_A = \{1,2\}$*, defenders* $N_D = \{3,4\}$*. Player 3 protects targets* t_1 *and* t_2*, while player 4 defends* t_3, t_4*. The capacities of attackers are* $a_1 = 0.5$*,* $a_2 = 0.8$*, and the defenders' capacities are* $d_3 = d_4 = 1$*. The efficiency functions are:*

$$e_1(x,y) = 0.7x^2(1-y^2)$$
$$e_2(x,y) = (1.4x - 0.7x^2)(y^2 - 2y + 1)$$
$$e_3(x,y) = (1.8x - 0.9x^2)(1-y^2)$$
$$e_4(x,y) = 0.9x^2(y^2 - 2y + 1)$$

Attackers in security games can attack any target, but each defender can only protect targets within their jurisdiction. The interactions can be modelled as a bipartite graph of pairwise general-sum games over targets to which the players allocate resources up to their total capacity. Each target is associated with a resiliency function e_t*. The total utility of a defender* j *is*

$$u_j(x) = \sum_{i \in N_A} \sum_{t \in T_j} e_t(x_i, x_j),$$

that is, the sum of resiliency evaluations over all attackers N_A *and all targets within their jurisdiction* T_j*; the total utility of an attacker* i *is*

$$u_i(x) = a_i - \sum_{j \in N_D} \sum_{t \in T_j} e_t(x_i, x_j).$$

Thus the games are globally constant sum.

Our algorithm returned the following partially-mixed strategy profile at order 6:

$$\mu_1^\star \approx \delta_{(0.29,0,0,0.21)},$$
$$\mu_2^\star \approx \delta_{(0.36,0.13,0.04,0.24)},$$
$$\mu_3^\star \approx 0.46\delta_{(0.33,0.67)} + 0.54\delta_{(0.69,0.31)},$$
$$\mu_4^\star \approx 0.02\delta_{(0.46,0.54)} + 0.06\delta_{(0.08,0.92)} + 0.92\delta_{(0.91,0.09)}$$

The payoffs to each player are $\mathbf{w} \approx (0.1, 0.3, -0.17, -0.23)$*. While the payoffs converged at order 4 already and took only 3 seconds to solve, the corresponding strategies could not be extracted until order 6, which took 6 minutes to solve. Due to the high dimensionality of the utility functions and the fast growth of the hierarchy, the resulting SDP contained over 25 thousand constraints.*

1.3. Epsilon Equilibria in Generic Games

In this section we present an algorithm that is much more widely applicable at the cost of relaxing the requirement on the exactness of solutions from the previous section. We focus on the computation and approximation of mixed strategy equilibria for the whole class of multiplayer general-sum continuous games. To solve such generic games, we vastly extend the scope of applicability of the double oracle algorithm, initially designed and proved to converge only for finite two-player zero-sum games. Specifically, we propose an iterative strategy generation technique, which splits the original problem into the master problem with only a finite subset of strategies being considered, and the subproblem in which an oracle finds the best response of each player. This simple method is guaranteed to recover an approximate equilibrium in finitely many iterations.

Further, we argue that the Wasserstein distance (the earth mover's distance) represents a very natural metric on the space of mixed strategies. Our main result is the proof of convergence of this algorithm in the Wasserstein distance to an equilibrium of the original continuous game.

There are relatively few methods for computing or approximating equilibria of continuous games, which narrows down the continuous games for which the exact solution can be found and used as a benchmark in our experiments. We demonstrate the computational performance of our method on selected examples of games appearing in the literature and on randomly generated games.

The double oracle algorithm [39] was extended from finite games and proved to converge for all two-player zero-sum continuous games [1]. The algorithm is based on the iterative solution of finite subgames and their subsequent extension with best responses. Despite its simplicity, this method was successfully adapted to large extensive-form games [7], Bayesian games [38], and security domains with complex policy spaces [67].

1.3.1. Subgames and Epsilon-Equilibria

Consider a game $\mathcal{G} = \langle N, (X_i)_{i\in N}, (u_i)_{i\in N}\rangle$ and nonempty compact subsets $Y_i \subseteq X_i$ for each player i from the player set N. When each u_i is restricted to $Y_1 \times \cdots \times Y_n$, the continuous game $\langle N, (Y_i)_{i\in N}, (u_i)_{i\in N}\rangle$ is called the *subgame* of $\mathcal{G}$.

The *support* of a mixed strategy μ is the compact set defined by

$$\operatorname{spt}\mu = \bigcap\{K \subseteq X \mid K \text{ compact}, \mu(K) = 1\}.$$

In this section, we will construct mixed strategies with finite supports. The support of Dirac measure δ_x is the singleton $\operatorname{spt}\delta_x = \{x\}$, where $x \in X_i$. In general, when the support $\operatorname{spt}\mu$ of a mixed strategy μ is finite, it means that $\mu(x) > 0$ for all $x \in \operatorname{spt}\mu$ and $\sum_{x\in\operatorname{spt}\mu}\mu(x) = 1$. For clarity, we emphasize that a finitely-supported mixed strategy μ_i of player i should be interpreted as the function $\mu : X_i \to [0, 1]$ vanishing outside $\operatorname{spt}\mu_i$, and not as a vector of probabilities with a fixed dimension. This is because only the former viewpoint enables us to consider the distance between *any* pair of pure strategies in X_i, which makes it possible to compute a distance between mixed strategies with *arbitrary* supports.

Proposition 1. *For each player $i \in N$ and any mixed strategy profile $\boldsymbol{\mu}_{-i} \in \mathbf{M}_{-i}$, there exists a pure strategy $x_i \in X_i$ such that*

$$\max_{p \in M_i} U_i(p, \boldsymbol{\mu}_{-i}) = \max_{x \in X_i} U_i(x, \boldsymbol{\mu}_{-i}) = U_i(x_i, \boldsymbol{\mu}_i).$$

Glicksberg's theorem (1) states that every continuous game has an equilibrium. The following useful characterization is a consequence of Proposition 1: A profile $\boldsymbol{\mu}^*$ is an equilibrium if, and only if,

$$U_i(x_i, \boldsymbol{\mu}^*_{-i}) \leq U_i(\boldsymbol{\mu}^*), \qquad \text{for every } i \in N \text{ and all } x_i \in X_i.$$

For any $\epsilon \geq 0$, an ϵ-*equilibrium* is a mixed strategy profile $\boldsymbol{\mu}^* \in \mathbf{M}$ such that

$$U_i(x_i, \boldsymbol{\mu}^*_{-i}) - U_i(\boldsymbol{\mu}^*) \leq \epsilon, \qquad \text{for every } i \in N \text{ and } x_i \in X_i.$$

This implies that for every $p_i \in M_i$, the inequality $U_i(p_i, \boldsymbol{\mu}^*_{-i}) - U_i(\boldsymbol{\mu}^*) \leq \epsilon$ also holds.

We introduce the vector notation

$$\mathbf{U}(\boldsymbol{\mu}) = (U_1(\boldsymbol{\mu}), \dots, U_n(\boldsymbol{\mu})).$$

Furthermore, we define

$$\mathbf{U}(\mathbf{x}, \boldsymbol{\mu}) = (U_1(x_1, \boldsymbol{\mu}_{-1}), \dots, U_n(x_n, \boldsymbol{\mu}_{-n})),$$

for any $\mathbf{x} \in \mathbf{X}$ and $\boldsymbol{\mu} \in \mathbf{M}$. Let $\boldsymbol{\epsilon}$ be the vector $(\epsilon, \dots, \epsilon)$. Using this vector notation, a mixed strategy profile $\boldsymbol{\mu}^* \in \mathbf{M}$ is an ϵ-equilibrium if, and only if,

$$\mathbf{U}(\mathbf{x}, \boldsymbol{\mu}^*) - \mathbf{U}(\boldsymbol{\mu}^*) \leq \boldsymbol{\epsilon}, \qquad \forall \mathbf{x} \in \mathbf{X}.$$

1.3.2. Convergence of Mixed Strategies

We consider an arbitrary metric ρ_i on the compact strategy space $X_i \subseteq \mathbb{R}^{m_i}$ of each player $i \in N$. This enables us to quantify a distance between pure strategies $x, y \in X_i$ by the number $\rho_i(x, y) \geq 0$. Consequently, we can define the Wasserstein distance d_W on M_i, which is compatible with the metric of the underlying strategy space X_i in the sense that

$$d_W(\delta_x, \delta_y) = \rho_i(x, y), \qquad \forall x, y \in X_i.$$

The preservation of distance from X_i to M_i is a very natural property, since the space of pure strategies X_i is embedded in M_i via the correspondence $x \mapsto \delta_x$ mapping the pure strategy x to the Dirac measure δ_x.

The Wasserstein distance originated from optimal transport theory. It is nowadays highly instrumental in solving many problems of computer science. Specifically, the *Wasserstein distance* of mixed strategies $p, q \in M_i$ is

$$d_W(p, q) = \inf_{\mu} \int_{X_i^2} \rho_i(x, y) \, \mathrm{d}\mu(x, y),$$

where the infimum is over all Borel probability measures μ on X_i^2 whose one-dimensional marginals are p and q:

$$\mu(A \times X_i) = p(A),$$
$$\mu(X_i \times A) = q(A),$$

for all Borel subsets $A \subseteq X_i$.

The dependence of d_W on the metric ρ_i is understood. The function d_W is a metric on M_i. Since X_i is compact, it has necessarily bounded diameter. This implies that the convergence in (M_i, d_W) coincides with the weak convergence [48, Corollary 2.2.2]. Specifically, the following two assertions are equivalent for any sequence (p^j) in M_i:

1. (p^j) converges to p in (M_i, d_W).

2. (p^j) *weakly converges* to p, which means by the definition that

$$\lim_j \int_{X_i} f \, \mathrm{d}p^j = \int_{X_i} f \, \mathrm{d}p,$$

   for every continuous function $f : X_i \to \mathbb{R}$.

The metric space (M_i, d_W) is compact by [48, Proposition 2.2.3]. Consequently, the joint strategy space $\mathbf{M}$ is compact in a product metric as well, and any sequence $(\boldsymbol{\mu}^j)$ in $\mathbf{M}$ has an accumulation point. Equivalently, the previous assertion can be formulated as follows.

Proposition 2. *Any sequence of mixed strategy profiles $(\boldsymbol{\mu}^j)$ in $\mathbf{M}$ contains a weakly convergent subsequence.*

The function U_i is continuous on $\mathbf{M}$ by the definition of weak convergence. This implies that if $(\boldsymbol{\mu}^j)$ weakly converges to $\boldsymbol{\mu}$ in $\mathbf{M}$, then the corresponding values of expected utility goes to $U_i(\boldsymbol{\mu})$, that is, $\lim_j U_i(\boldsymbol{\mu}^j) = U_i(\boldsymbol{\mu})$. By compactness of M_i and continuity of U_i, all maxima and maximizers appearing in the paper exist. This implies that the optimal value of utility function in response to the mixed strategies of other players is attained for some pure strategy. Proposition 1 is the precise formulation of this useful fact.

In the rest of this section, we will discuss the problem of computing and approximating the Wasserstein distance $d_W(p, q)$ of any pair of mixed strategies $p, q \in M_i$ for player i. In general, this is a difficult infinite-dimensional optimization problem [51]. In our setting it suffices to evaluate $d_W(p, q)$ only for mixed strategies with finite supports. In particular, it follows immediately from the definition of d_W that

$$d_W(p, \delta_y) = \sum_{x \in \operatorname{spt} p} \rho_i(x, y) p(x),$$

for every finitely-supported mixed strategy $p \in M_i$ and any $y \in X_i$. If mixed strategies p and

q have finite supports, then computing $d_W(p, q)$ becomes the linear programming problem

$$\begin{aligned}
\min_{\mu(x,y)} \quad & \sum_{x \in \operatorname{spt} p} \sum_{y \in \operatorname{spt} q} \rho_i(x, y)\mu(x, y) \\
& \mu(x, y) \geq 0 \qquad \forall (x, y) \in \operatorname{spt} p \times \operatorname{spt} q \\
& \sum_{x \in \operatorname{spt} p} \sum_{y \in \operatorname{spt} q} \mu(x, y) = 1, \\
& \sum_{y \in \operatorname{spt} q} \mu(x, y) = p(x) \qquad \forall x \in \operatorname{spt} p \\
& \sum_{x \in \operatorname{spt} p} \mu(x, y) = q(y) \qquad \forall y \in \operatorname{spt} q
\end{aligned} \tag{1.10}$$

with variables $\mu(x, y)$ indexed by $(x, y) \in \operatorname{spt} p \times \operatorname{spt} q$.

1.3.3. The Multiple Oracle algorithm and its Correctness

We propose Algorithm 1 as an iterative strategy-generation technique for (approximately) solving any continuous game $\mathcal{G} = \langle N, (X_i)_{i \in N}, (u_i)_{i \in N} \rangle$. We recall that the *best response set* of player $i \in N$ with respect to a mixed strategy profile $\boldsymbol{\mu}_{-i} \in \mathbf{M}_{-i}$ is

$$\beta_i(\boldsymbol{\mu}_{-i}) = \operatorname*{arg\,max}_{x \in X_i} U_i(x, \boldsymbol{\mu}_{-i}).$$

The set $\beta_i(\boldsymbol{\mu}_{-i})$ is always nonempty by Proposition 1. We assume that every player uses an oracle to recover at least one best response strategy, which means that the player is able to solve the corresponding optimization problem to global optimality. In Section 1.3.6 we will see the instances of games for which this is possible.

Algorithm 1 The Multiple Oracle algorithm

```
function ITERATE(𝒢, X)
    Find an equilibrium μ of the subgame ⟨N, X, (u_i)_{i∈N}⟩
    ∀i ∈ N : Pick a best response x_i ∈ β_i(μ_{-i})
    ∀i ∈ N : Extend strategy set Y_i ← X_i ∪ {x_i}
    return Y, x, μ
end function
function MULTIPLE-ORACLE(𝒢, X, ε)
    repeat
        X, x, μ ← iterate(𝒢, X)
    until U(x, μ_{-i}) − U(μ) ≤ ε
    return μ
end function
```

Algorithm 1 proceeds as follows: In every iteration j, finite strategy sets X_i^j are constructed for each player $i \in N$ and the corresponding finite subgame of $\mathcal{G}$ is solved. Let $\boldsymbol{\mu}^j$ be its equilibrium. Then, an arbitrary best response strategy x_i^{j+1} with respect to $\boldsymbol{\mu}_{-i}^j$ is added to each strategy set X_i^j. Those steps are repeated until

$$U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j) \leq \epsilon \qquad \forall i \in N. \tag{1.11}$$

First we discuss basic properties of the algorithm. At each step j, we have the inequality

$$\mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) \geq \mathbf{U}(\boldsymbol{\mu}^j). \tag{1.12}$$

Indeed, for each player $i \in N$, we get

$$U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) = \max_{x \in X_i} U_i(x, \boldsymbol{\mu}_{-i}^j) \geq \max_{x \in X_i^j} U_i(x, \boldsymbol{\mu}_{-i}^j) = U_i(\boldsymbol{\mu}^j).$$

We note that the stopping condition of the double oracle algorithm for two-player zero-sum continuous games [1] is necessarily different from (1.11). Namely the former condition, which is tailored to the zero-sum games, is

$$U_1(x_1^{j+1}, \mu_2^j) - U_1(x_2^{j+1}, \mu_1^j) \leq \epsilon. \tag{1.13}$$

When $\mathcal{G}$ is a two-player zero-sum continuous game, it follows immediately from (1.12) that (1.13) implies (1.11).

We prove correctness of Algorithm 1 — the eventual output $\boldsymbol{\mu}^j$ is an ϵ-equilibrium of the original game $\mathcal{G}$.

Lemma 1. *The strategy profile $\boldsymbol{\mu}^j$ is an ϵ-equilibrium of $\mathcal{G}$, whenever Algorithm 1 terminates at step j.*

Proof. Let $\mathbf{x} \in \mathbf{X}$. We get

$$\mathbf{U}(\mathbf{x}, \boldsymbol{\mu}^j) - \mathbf{U}(\boldsymbol{\mu}^j) = \mathbf{U}(\mathbf{x}, \boldsymbol{\mu}^j) - \mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) + \mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) - \mathbf{U}(\boldsymbol{\mu}^j).$$

Then $\mathbf{U}(\mathbf{x}, \boldsymbol{\mu}^j) - \mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) \leq \mathbf{0}$, since $\mathbf{x}^{j+1}$ is the profile of best response strategies. Consequently, we obtain

$$\mathbf{U}(\mathbf{x}, \boldsymbol{\mu}^j) - \mathbf{U}(\boldsymbol{\mu}^j) \leq \mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) - \mathbf{U}(\boldsymbol{\mu}^j) \leq \boldsymbol{\epsilon},$$

where the last inequality is just the terminating condition (1.11). Therefore, $\boldsymbol{\mu}^j$ is an ϵ-equilibrium of $\mathcal{G}$. □

If the set $\mathbf{X}^j = X_1^j \times \cdots \times X_n^j$ cannot be further inflated, Algorithm 1 terminates.

Lemma 2. *Assume that $\mathbf{X}^j = \mathbf{X}^{j+1}$ at step j of Algorithm 1. Then*

$$\mathbf{U}(\mathbf{x}^{j+1}, \boldsymbol{\mu}^j) = \mathbf{U}(\boldsymbol{\mu}^j).$$

Proof. The condition $\mathbf{X}^j = \mathbf{X}^{j+1}$ implies $\mathbf{x}^{j+1} \in \mathbf{X}^j$. For each player $i \in N$,

$$U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) = \max_{x \in X_i^j} U_i(x, \boldsymbol{\mu}_{-i}^j) = U_i(\boldsymbol{\mu}^j).$$

□

The original double oracle algorithm for two-player zero-sum finite games [39] makes it possible to find exact equilibria. The following proposition generalizes this result to the setting of multiplayer general-sum finite games.

Proposition 3. *If $\mathcal{G}$ is a finite game and $\epsilon = 0$, then Algorithm 1 recovers an equilibrium of $\mathcal{G}$ in finitely-many steps.*

Proof. Let $\mathcal{G}$ be finite and $\epsilon = 0$. Since each X_i is finite, there exists an iteration j in which $\mathbf{X}^{j+1} = \mathbf{X}^j$. Then the terminating condition of Algorithm 1 is satisfied with $\epsilon = 0$ (Lemma 2) and $\boldsymbol{\mu}^j$ is an equilibrium of $\mathcal{G}$ (Lemma 1). □

Theorems 3 and 4 extend the convergence theorem from [1] and provides an alternative proof of Glicksberg's theorem [22] about existence of equilibria in continuous games. Indeed, the sequence $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \dots$ has at least one accumulation point by Proposition 2. Further, the consequence of Theorem 4 is that a finitely-supported ϵ-equilibrium of $\mathcal{G}$ can always be found in finitely-many steps, although game $\mathcal{G}$ may have no finitely-supported equilibria.

Theorem 3. *Let $\mathcal{G}$ be a continuous game and $\epsilon = 0$. If Algorithm 1 stops at step j, then $\boldsymbol{\mu}^j$ is an equilibrium of $\mathcal{G}$. Otherwise, any accumulation point of $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \dots$ is an equilibrium of $\mathcal{G}$.*

Proof. If Algorithm 1 terminates at step j, then Lemma 1 implies that $\boldsymbol{\mu}^j$ is an equilibrium of $\mathcal{G}$. In the opposite case, the algorithm generates a sequence of mixed strategy profiles $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \dots$ Consider any weakly convergent subsequence of this sequence — at least one such subsequence exists by Proposition 2. Without loss of generality, such a subsequence will be denoted by the same indices as the original sequence. Therefore, for each player $i \in N$, there exists some $\mu_i^\star \in M_i$, such that the sequence $\mu_i^1, \mu_i^2, \dots$ weakly converges to $\mu_i^\star$. We need to show that $\boldsymbol{\mu}^\star = (\mu_1^\star, \dots, \mu_n^\star) \in \mathbf{M}$ is an equilibrium of $\mathcal{G}$.

For all $i \in N$, define

$$Y_i = \bigcup_{j=1}^{\infty} X_i^j.$$

First, assume that $x \in Y_i$. Then there exists j_0 with $x \in X_i^j$ for each $j \geq j_0$. Hence the inequality $U_i(\boldsymbol{\mu}^j) \geq U_i(x, \boldsymbol{\mu}_{-i}^j)$, for each $j \geq j_0$, since $\boldsymbol{\mu}^j$ is an equilibrium of the corresponding finite subgame. Therefore,

$$U_i(\boldsymbol{\mu}^\star) = \lim_j U_i(\boldsymbol{\mu}^j) \geq \lim_j U_i(x, \boldsymbol{\mu}_{-i}^j) = U_i(x, \boldsymbol{\mu}_{-i}^\star).$$

Further, by continuity of U_i,

$$U_i(\boldsymbol{\mu}^\star) \geq U_i(x, \boldsymbol{\mu}_{-i}^\star) \quad \text{for all } x \in \mathrm{cl}(Y_i). \tag{1.14}$$

Now, consider an arbitrary $x \in X_i \setminus \mathrm{cl}(Y_i)$. The definition of x_i^{j+1} yields $U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) \geq U_i(x, \boldsymbol{\mu}_{-i}^j)$ for each j. This implies, by continuity,

$$\lim_j U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^\star) \geq \lim_j U_i(x, \boldsymbol{\mu}_{-i}^j) = U_i(x, \boldsymbol{\mu}_{-i}^\star). \tag{1.15}$$

Since $x_i^{j+1} \in X_i^{j+1}$, compactness of X_i provides a convergent subsequence (denoted by the same indices) such that $x' = \lim_j x_i^j \in \mathrm{cl}(Y_i)$. Then (1.14) gives

$$U_i(\boldsymbol{\mu}^\star) \geq U_i(x', \boldsymbol{\mu}_{-i}^\star) = \lim_j U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^\star). \tag{1.16}$$

Combining (1.15) and (1.16) shows that $U_i(\boldsymbol{\mu}^\star) \geq U_i(x, \boldsymbol{\mu}_{-i}^\star)$. □

Theorem 4. *Let $\mathcal{G}$ be a continuous game and $\epsilon > 0$. Then Algorithm 1 terminates at some step j and $\boldsymbol{\mu}^j$ is an ϵ-equilibrium of $\mathcal{G}$.*

Proof. If Algorithm 1 terminates at step j with a positive ϵ, then Lemma 1 implies that $\boldsymbol{\mu}^j$ is an ϵ-equilibrium of $\mathcal{G}$. Otherwise Algorithm 1 produces a sequence $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \ldots$ and we can repeat the analysis as in Item 1 for convergent subsequences of $(\boldsymbol{\mu}^j)$ and $(\mathbf{x}^j)$, which are denoted by the same indices. Define $x' = \lim_j x_i^j$. Then

$$U_i(\boldsymbol{\mu}^\star) \geq U_i(x', \boldsymbol{\mu}_{-i}^\star) = \lim_j U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j). \tag{1.17}$$

At every step j we have $U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) \geq U_i(\boldsymbol{\mu}^j)$ for each $i \in N$ by (1.12). For this reason $\lim_j U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) \geq U_i(\boldsymbol{\mu}^\star)$. Putting together the last inequality with (1.17), we get

$$\lim_j (U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j)) = 0 \tag{1.18}$$

for each $i \in N$. This equality means that Algorithm 1 stops at some step j and $\boldsymbol{\mu}^j$ is an ϵ-equilibrium by Lemma 1. □

Algorithm 1 generates the sequence of equilibria $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \ldots$ in increasingly larger subgames of $\mathcal{G}$. The sequence itself may fail to converge weakly in $\mathbf{M}$ even for a two-player zero-sum continuous game; see Example 1 from [1]. In fact, Theorems 3 and 4 guarantee convergence only to an accumulation point. We recall that this is a typical feature of some globally convergent methods not only in infinite-dimensional spaces [27, Theorem 2.2], but also in Euclidean spaces. For example, a gradient method generates the sequence such that only its accumulation points are guaranteed to be stationary points [6, Proposition 1.2.1]. One necessary condition for the weak convergence of $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \ldots$ is easy to formulate using the stopping condition of Algorithm 1.

Proposition 4. *If the sequence $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \ldots$ generated by Algorithm 1 converges weakly to an equilibrium $\boldsymbol{\mu}^\star$, then $\lim_j \big(U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j)\big) = 0$, for each $i \in N$.*

Proof. Using (1.12) and the triangle inequality,

$$\begin{aligned} U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j) &= \left| U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j) \right| \\ &\leq \left| U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^\star) \right| + \left| U_i(\boldsymbol{\mu}^\star) - U_i(\boldsymbol{\mu}^j) \right|. \end{aligned}$$

As $j \to \infty$, the first summand goes to zero by (1.18) and the second by the assumption. Hence the conclusion. □

In our numerical experiments (see Section 1.3.6), we compute the difference

$$U_i(x_i^{j+1}, \boldsymbol{\mu}_{-i}^j) - U_i(\boldsymbol{\mu}^j) \tag{1.19}$$

at each step j and check if such differences are diminishing with j increasing. This provides a simple heuristics to detect the quality of approximation and convergence. Another option is to calculate the Wasserstein distance

$$d_W(\boldsymbol{\mu}^j, \boldsymbol{\mu}^{j+1}), \tag{1.20}$$

which can be done with a linear program (1.10). If the sequence $\boldsymbol{\mu}^1, \boldsymbol{\mu}^2, \ldots$ converges weakly, then $d_W(\boldsymbol{\mu}^j, \boldsymbol{\mu}^{j+1}) \rightarrow 0$. We include the values (1.20) in the results of some numerical experiments and observe that they are decreasing to zero quickly. Figure 1.3.6 shows, that neither (1.19) nor (1.20) are monotone sequences.

Algorithm 1 is a meta-algorithm, which is parameterized by

1. the algorithm for computing equilibria of subgames (the *master problem*) and

2. the optimization method for computing the best response (the *oracles*).

We detail this setup for each example in the next section.

The choice of computational methods should reflect the properties of a continuous game, since the efficiency of methods for solving the master problem and subproblem is the decisive factor for the overall performance and precision of Algorithm 1. For example, polymatrix games are solvable in polynomial time [10], whereas finding even an approximate Nash equilibrium of a finite general-sum game is a very hard problem [14]. As for the solution of the oracles, the best response computation can be based on global solvers for special classes of utility functions.

1.3.4. Implementation of the Master Problem

Finding NE in zero-sum finite games reduces to either the classical linear program, or its aforementioned extension by Cai et al. [10] in case of polymatrix games. The implementation of the master problem corresponding to general-sum matrix sub-games is more complex. Various methods exist to solve general-sum games, including the global Newton method, simplical subdivision, or even straightforward strategy enumeration. These methods, however, may have unpredictable runtimes and do not always guarantee convergence.

In our experiments on general-sum matrix games, we use the path following algorithm implemented by the *gambit-logit* utility which is a part of the Gambit library, along with a version of a multilinear formulation of NE (1.21) described by Sturmfels [61] and improved by Fischer and Gupte [19]. We also use the classical linear programming implementation where appropriate.

Continuing with the established notation, let $w_i \in \mathbb{R}$ be the bound on player i's utility, and let $\boldsymbol{\mu}$ be probability simplices defining the mixed strategies of players. The multilinear feasibility program in $\mathbf{w}$ and $\boldsymbol{\mu}$ for finding NE in general-sum games is

$$\begin{aligned} &\sum_{i\in N} U_i(\boldsymbol{\mu}) \geq \sum_{i\in N} w_i && \\ &U_i(x, \boldsymbol{\mu}_{-i}) \leq w_i && \forall i \in N, x \in X_i \\ &\sum_{x\in X_i} \mu_i(x) = 1 && \forall i \in N \\ &0 \leq \mu_i(x) \leq 1 && \forall i \in N, x \in X_i \end{aligned} \tag{1.21}$$

As the utility functions are defined by tensors and strategy spaces are finite, the definitions of expected utility simplify into weighted sums. Specifically, the best response function $U_i(x, \boldsymbol{\mu}_{-i})$ reduces to the vector

$$\sum_{\mathbf{x}_{-i}\in\mathbf{X}_{-i}} U_i[x, \mathbf{x}_{-i}] \prod_{j\in N_{-i}} \mu_j[x_j];$$

while the expected utility $U_i(\boldsymbol{\mu})$ becomes

$$\sum_{\mathbf{x}\in\mathbf{X}} U_i[\mathbf{x}] \prod_{i\in N} \mu_i[x_i].$$

As mentioned in **??**, algebraic functions can be decomposed into binary operations for which there exist known tight convex relaxations. Specifically, the multilinear program 1.21 can be reformulated as a non-convex quadratically-constrained quadratic program, which can then be solved by spatial Branch and Bound solvers.

An advantage of the multilinear program, is that it can be modified to search for specific solutions. Constraining the search to symmetric equilibria can be done trivially by adding variables μ_0 and equating $\mu_0 = \mu_i \ \forall i \in N$. We use this modification to exploit the symmetry of Blotto games and speed up the search in experiment 2. Additionally, as 1.21 is a feasibility program, it can be trivially modified to search for equilibria with the smallest ℓ_0 norm.

The 1.21 formulation worked acceptably well with *Gurobi*, and frequently outperformed other methods. Fig. 1.2 demonstrates that we could not replicate the consistency claimed by Fischer and Gupte [19]. We note that in our experience, random matrices tend to be poor anologues for the subgames generated by the *Double Oracle* algorithm. To provide a more realistic benchmark we discretized the strategy spaces of two different test functions in Fig. 1.3, where the multilinear formulation was more consistent; however, in actual runs of the algorithm, the master-problem still occasionally times out when impelemnted in this way. It appears that consistent performance relies on optimizations specific to the *Baron* solver, which we could not run sufficiently extensive benchmarks due to a lack of licence.

plotting error: timed-out results should be counted as the maximum value not as missing values, otherwise the averages don't sort properly

Finally, solvers may be able to take advantage of the structure of a game formulated as the multilinear program. Fig. 1.4 shows that *Gurobi* is significantly faster on zero-sum compared to general-sum games. The performance of the *gambit-logit* utility, on the other hand, is consistent across our tests.

In our testing, the *Gambit logit* utility was the fastest on average and most reliable method for solving finite general-sum games, which agrees with the benchmarks done by Fischer and Gupte [19], where it was the second fastest method.

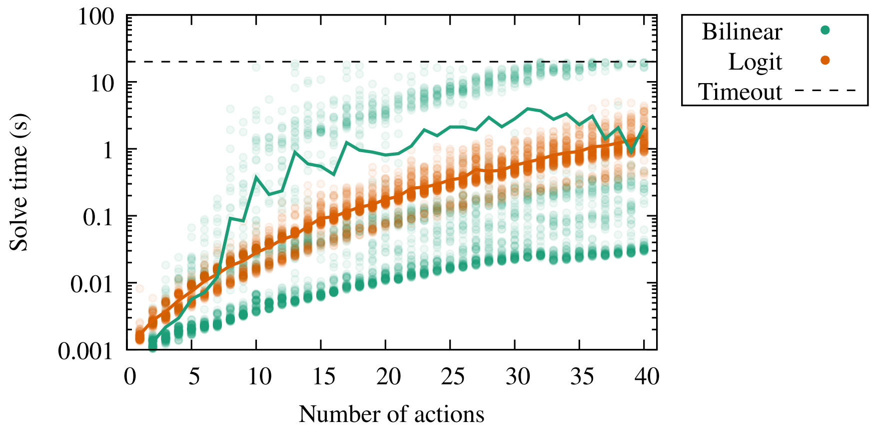


Figure 1.2.: Runtime comparison between the bilinear formulation and the *gambit-logit* algorithm on two-player general-sum matrix games with normally distributed payoffs. Each algorithm was executed 100 times per game size with a 20-second time limit. Mean runtimes of successful runs are overlaid as thick lines. Although the bilinear program exhibits lower median runtimes (shown as darkest plot regions) compared to *gambit-logit* with default settings, its runtime has a multimodal distribution with a significantly higher standard deviation (5.15 s vs. 0.22 s for 30 × 30 games) leading to frequent timeouts. The *gambit-logit* program always finished successfully, while the success rate of the bilinear formulation was steadily decreasing, completing 89% of the 30-action games and only 59% of 40-action games. Mean runtimes on 20-action games were: 0.81 s (bilinear) and 0.18 s (*gambit-logit*).

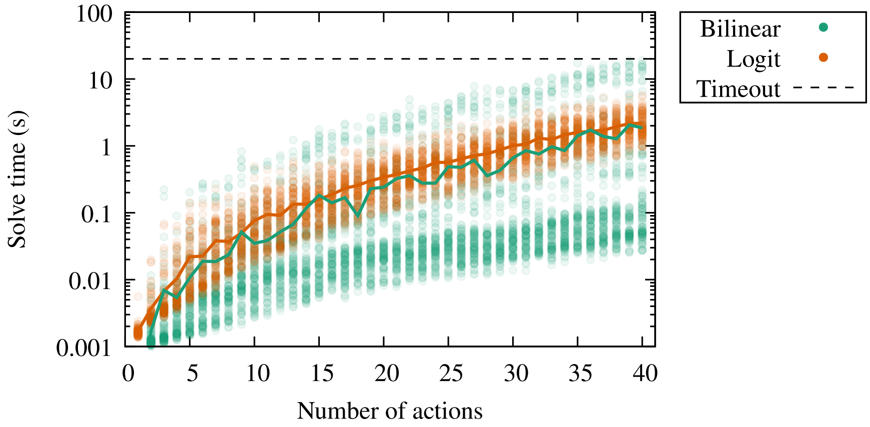


Figure 1.3.: Runtime comparison between the bilinear formulation and the *gambit-logit* algorithm on two-player general-sum matrix games. Each algorithm was executed 100 times per game size with a 20-second time limit. Solid lines represent mean runtimes. Payoff matrices were generated by evaluating the Michalewicz function for Player 1 and McCormick function for Player 2 on a Cartesian product of randomly sampled actions uniformly distributed over $[0, \pi]$. This test should provide a more realistic benchmark for the Double Oracle master-problem compared to normally-distributed random matrices.

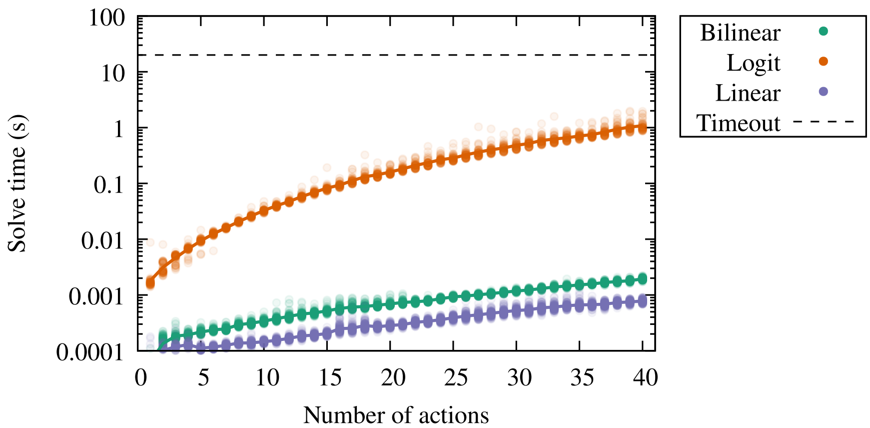


Figure 1.4.: Runtime comparison of the bilinear program, linear program, and the *gambit-logit* algorithm for solving two-player zero-sum matrix games with normally distributed payoffs. Each algorithm was executed 100 times per game size with a 20-second time limit. Thick lines indicate mean runtimes of successful executions. For 20-action games, mean runtimes were: 0.6 ms (bilinear program), 0.02 ms (linear program), and 159.7 ms (*gambit-logit*).

1.3.5. Implementation of the Oracles

In games with polynomial utility functions, the best response oracles can employ both methods of the methods of global optimization described in Chapter **??**. In the rest of the games we use the spatial Branch and Bound approach as the best response oracle. From our experience, local solvers can work sufficiently well as oracles in some games, although no convergence guarantees then apply.

this belongs elsewhere (to the "this is a metaalgorithm parametrized by master and oracle...".... ok now it's here)

blah...The examples in our collections have utilities specified in terms of sums and products of functions such as logarithms and square roots. All such functions should be accepted either directly by global solvers or through external modeling libraries such as AMPL.

merge this properly

We show a comparison of Oracle implementations based on non-linear solvers. While we have succesfully used local-solvers as oracles, it is important to remember that convergence to equilibria is no longer guaranteed. Regardless, we compare their performance in some of the following experiments.

The experiments are split into three groups, random polynomials, non-polynomial test functions, and a practical experiment where we solve a where we run a fixed number of iterations of Double Oracle on the Colonel Blotto game.

polynomials blah

non-polynomial test functions blah

The next experiment is a measurement of the runtime of different solvers in practice over a course of ten iterations of double oracle in a game of General Blotto. The utility functions in *General Blotto* games are sums of non-linearly weighted margins of victory over sets of fronts. We chose a game with three isolated fronts defined by the utility $u_1(x_1, x_2) = \sum_{j=1}^{m} f\left(x_{1j} - x_{2j}\right)$ with a non-differentiable weighing function $f(z) = \operatorname{sgn}(z)\,|z|^{0.5}$. This game exhibits very slow convergence with the Double Oracle algorithm.

1.3.6. Numerical Experiments

We will compare the results obtained by specialized methods and detail the setup for each example in the upcoming section. Later we switch to a generic method for each subproblem and show the versitility of the Multiple Oracle method.

moved here, just check and inline properly

We show the performance that can be expected from Multiple Oracle on various important examples from the literature. In this section we present only the examples which we thought demonstrate interesting behaviors, and also special classes of games for which efficient oracles and master-problems exist. In appendix **??**, we follow with a more comprehensive evaluation of a general implementation of the algorithm based on nonlinear solvers and finite general-sum games.

We initialize the algorithm by first selecting a feasible point[1] and then allowing each player to select a best response to the feasible point. The algorithm then continues with standard Multiple Oracle iterations: by computing an equilibrium of the finite subgame and selecting additional best responses.

We plot the progress of the largest possible gain from a profitable deviation (1.19, ϵ) at each iterations, along with the Wasserstein distance (1.20) between mixed strategies from consecutive iterations.

Example 2 (General Blotto). *The utility functions in* General Blotto *games are sums of non-linearly weighted margins of victory over sets of fronts. We chose a game with three isolated fronts defined by the utility* $u_1(x_1, x_2) = \sum_{j=1}^{m} f\left(x_{1j} - x_{2j}\right)$*, with the weighing function* $f(z) = \operatorname{sgn}(z)\,|z|^p$. *When* $p = 0.5$*, the utility function is non-differentiable and the game converges very slowly as shown in Fig. 1.5. Large gains are possible by winning a front by a small margin as the slope of the utility function around zero approaches infinity. After 140 iterations, the algorithm recovers the* ϵ*-NE with* $\epsilon = 0.006$ *shown in figure 1.6.*

Multilinear terms, square roots and sums can be formulated directly via the generic modelling languages JuMP, *and* AMPL*; however the sign function and absolute value are more challenging. Due to a lack of support and differentiability, all of the solvers:* Ipopt, Bonmin, Couenne, SCIP, *and* Gurobi *were unable to solve a direct formulation of the oracle problem. Instead, we formulated the oracle as the following mixed-integer quadratic program:*

$$
\begin{aligned}
\max_{b,z^+,z^-,x,u} \quad & \sum_{j=1}^{m} w_j \sum_{i=1}^{3} (1 - 2b_{ij}) u_{ij} & \\
b_{ij} &\in \{0, 1\} & \forall i, j \\
u_{ij}, x_{ij}, z_{ij}^+, z_{ij}^- &\in [0, 1] & \forall i, j \\
z_{ij}^+ &\leq b_{ij} & \\
z_{ij}^- &\leq 1 - b_{ij} & \\
x_{ij} - c_{ij} &= z_{ij}^+ - z_{ij}^- & \\
u_{ij}^2 &= z_{ij}^+ + z_{ij}^- &
\end{aligned}
$$

where c_{ij} *is the allocation of the opponent to the ith front in the jth action out of m in the mixed strategy of the opponent weighted by* w_j*.*

The game is symmetric, allowing us to speed up the algorithm by restricting the search to symmetric equilibria. The master-problem was implemented using the bilinear formulation (1.21) *with the additional constraint that all strategies* μ_i *be equal. Only one oracle computation per iteration was necessary due to the symmetry. The runtime to reach 140 iterations was still around 8 minutes.*

[1] We did not see a consistent difference in runtime when selecting a point in the interior of the players' strategy spaces, compared to choosing an arbitrary point found heuristically by solvers.

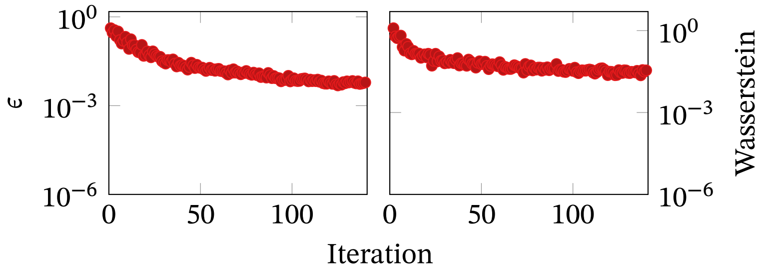


Figure 1.5.: Slow convergence of a continuous but non-differentiable *General Blotto* game valuing 3 isolated fronts with $p = 0.5$. Both players can still profitably deviate by 0.006 utility units after 140 iterations.

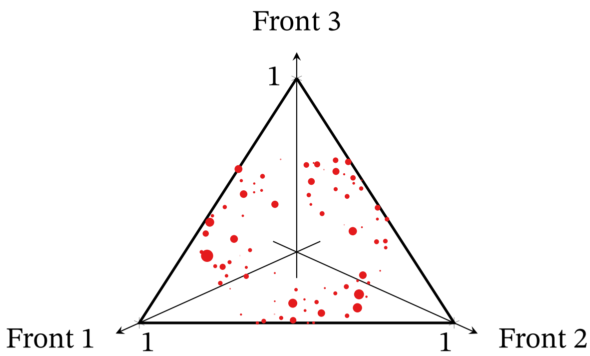


Figure 1.6.: The recovered symmetric strategy after 140 iterations drawn as circles on a simplex, with weight indicated by their size.

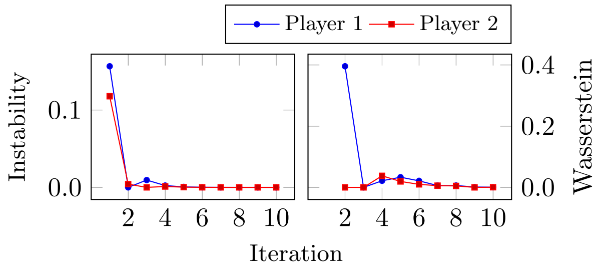


Figure 1.7.: Convergence in Example 39. After 10 iterations, our method finds pure strategies $x \approx 0.4$ and $y \approx 0.63$, resulting in payoffs $(-0.47, 0.47)$.

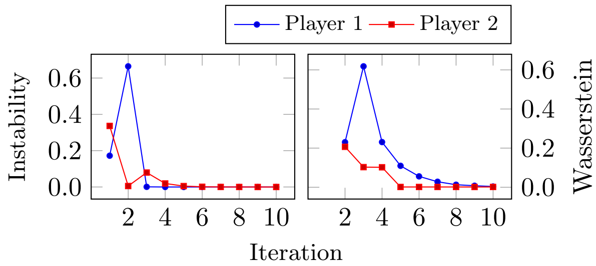


Figure 1.8.: Convergence in Example 42. Our method finds mixed strategies where $x \approx 0.11$ with probability 44.19% and $x \approx -1$ with probability 55.81% and $y \approx 0.72$ resulting in payoffs $(1.13, 1.81)$.

Example (Example 39)**.** *Consider a two-player zero-sum game with strategy sets* $[-1, 1]$ *and a utility function of the first player*

$$u(x, y) = 2xy^2 - x^2 - y.$$

As the generated subgames are all finite two-player zero-sum games, we can use linear programming to find their equilibria. The best response oracles were implemented using Lasserre hierarchies.

This example shows that neither the sequence of ϵ *nor the Wasserstein distances are monotone.*

Example (Example 42)**.** *The strategy set of each player is* $[-1, 1]$ *and utility functions are*

$$u_1(x, y) = -3x^2y^2 - 2x^3 + 3y^3 + 2xy - x,$$
$$u_2(x, y) = 2x^2y^2 + x^2y - 4y^3 - x^2 + 4y.$$

We computed equilibria in the general-sum matrix subgames using the multilinear program 1.21. The best response oracles were implemented using Lasserre hierarchies.

Example 3 (Random polymatrix games)**.** *Separable network games (polymatrix games) with finitely many strategies of each player can be solved in polynomial time [10] by linear programming. We use Algorithm 1 to compute equilibria of five-player polymatrix games defined by 20 by*

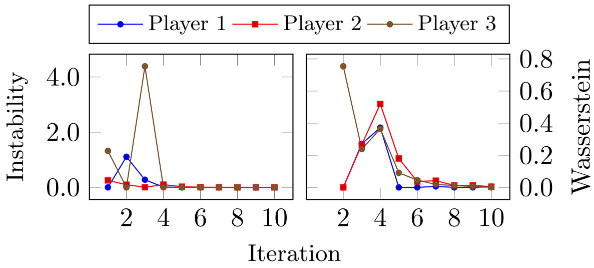


Figure 1.9.: Convergence of Example 48. The Multiple Oracle algorithm finds a solution in mixed strategies where $x_1 \approx -0.06$, x_2 is 0.35 with probability 0.708 and 0.36 with probability 0.292%, x_3 is 1.0 with probability 0.7211 and −1.0 with probability 0.2789; with the corresponding payoffs $(-1.23, 0.26, 0.97)$.

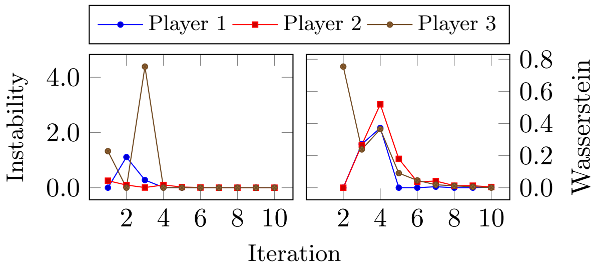


Figure 1.10.: Convergence of Example 49. TODO.

20 matrices. Specifically, we generated a random matrix for each edge in the network and then transposed and subtracted the matrix of utility functions to make the game globally zero sum. We remark that zero-sum polymatrix games are payoff-equivalent to the general polymatrix games by [10, Theorem 7]. In a test of 100 games, our algorithm found an ε-equilibrium with $\epsilon = 0.01$ *after* 17.69 *iterations in a mean time of* 0.29 *seconds.*

Example 4 (Separable Network Games)**.** *Thanks to the generality of the Multiple Oracle algorithm, we can also solve the polynomial network games 48 and 49, which were analyzed with the tools of polynomial optimization earlier in Section 1.2.3. We show their convergence in Figs. 1.9 and 1.10. The master problem was solved using the LP technique by Cai et al. [10, Linear Program 1], while the oracles were implemented using Lasserre hierarchies.*

rerun small network game and add runtime as a comparison for the generic algorithm

run large security game example as a comparison for the generic algorithm

A. List of Games

While there exist datasets, such as GAMUT, useful for testing algorithms designed for finite games, we found no established projects for continuous games. To address this, we present a compilation of parametrized games drawn from both recent and classical literature, which should provide a convincing and comprehensive evaluation of any game-solving algorithm.

Note that continuous games are inherently complex, as even apparently simple games may only have Nash equilibria supported on infinitely many points. A case in point are Examples 22 and 25 played on the $[0, 1]$ interval.

We chose to consistently present the utility function of player i denoted as u_i in the variables x_{ij}, where the first digit i denotes the player, while j is the variable index. None of the games that follow have more than ten players nor variables for any of the players.

A.1. General Blotto

The following examples are various settings for the classical two-player zero-sum game known as *General Blotto*. The utility functions are sums of non-linearly weighted margins of victory over fronts. Golman and Page [24] completely characterized the existence of pure equilibria for the following family of weighing functions:

$$f(z) = \operatorname{sgn}(z)\,|z|^p$$

parametrized by p, which is strictly positive for continuous games. The margins of victory are differences of products of the allocations to each front.

Example 5. *A two-player zero-sum game in $m \times m$ variables on the simplex*

$$x_i \in \Delta^{m-1} = \left\{ z \in \mathbb{R}^m \;\middle|\; z_j \geq 0,\ \sum_{j=1}^{m} z_j = 1 \right\} \quad \forall i \in \{1, 2\}.$$

The utility function is defined by

$$u_1(x_1, x_2) = \sum_{j=1}^{m} f\left(x_{1j} - x_{2j}\right)$$

where $f(x) = \operatorname{sign}(x)\,|x|^p$ and $p > 0$. This is a continuous version of General Blotto, *where all isolated fronts have equal value. When $p = 1$, every strategy is optimal; when $p > 1$, players may allocate all resources to any one front; no pure equilibrium exists for p less than one [24].*

Example 6. *A two-player zero-sum game in $m \times m$ variables on the simplex*

$$x_i \in \Delta^{m-1} \quad \forall i \in \{1, 2\}.$$

The utility function is defined by

$$u_1(x_1, x_2) = \sum_{i=1}^{m} \sum_{j=i+1}^{m} f\left(x_{1i}x_{1j} - x_{2i}x_{2j}\right) + \sum_{i=1}^{m} f\left(x_{1i} - x_{2i}\right)$$

where $f(x) = \operatorname{sign}(x)\,|x|^p$ and $p > 0$. This is a continuous version of General Blotto, *where all pairs of fronts have equal value. When $p = 1$, players should allocate $\frac{1}{m}$ to all fronts; otherwise, no pure equilibrium exists [24].*

Example 7. *A two-player zero-sum game in $m \times m$ variables on the simplex*

$$x_i \in \Delta^{m-1} \quad \forall i \in \{1, 2\}.$$

The utility function is defined by

$$u_1(x_1, x_2) = \sum_{\substack{I \subseteq \{1,\ldots,m\} \\ I \neq \emptyset}} f\left(\prod_{i \in I} x_{1i} - \prod_{i \in I} x_{2i}\right)$$

where $f(x) = \operatorname{sign}(x)\,|x|^p$ and $p > 0$. This is a continuous version of General Blotto, *where all sets of fronts have equal value. When $p = 1$ players should allocate $\frac{1}{m}$ to all fronts. No pure equilibrium exists for other values of p [24].*

A.2. Polynomial Saddle Point Games

The following examples are the saddle point problems of polynomials studied by Nie, Yang, and Zhou [45], which we have reinterpreted as two-player zero-sum polynomial games.

Example 8. *A zero-sum 2-player polynomial game in 3×3 variables on the simplex*

$$x_i \in \Delta^2 \quad \forall i \in \{1, 2\}.$$

The nonconcave-nonconvex utility function [45, Example 6.1 (i)] is

$$u_1(x_1, x_2) = -x_{11}x_{12} - x_{11}x_{23} - x_{12}x_{13} - x_{13}x_{21} - x_{21}x_{22} - x_{22}x_{23}$$

This game has the pure NE

$$x_1^\star \approx (0, 1, 0),\ x_2^\star \approx (0.25, 0.5, 0.25).$$

Example 9. *A zero-sum 2-player polynomial game in* 3×3 *variables on the simplex*

$$x_i \in \Delta^2 \quad \forall i \in \{1,2\}.$$

The nonconcave-nonconvex utility function [45, Example 6.1 (ii)] is

$$\begin{aligned} u_1(x_1,x_2) = &- x_{11}^3 - x_{12}^3 + x_{13}^3 + x_{21}^3 + x_{22}^3 - x_{23}^3 \\ &- x_{13}x_{21}x_{22}(x_{21}+x_{22}) - x_{12}x_{21}x_{23}(x_{21}+x_{23}) - x_{11}x_{22}x_{23}(x_{22}+x_{23}) \end{aligned}$$

This game has the pure NE

$$x_1^\star \approx (0,0,1),\ x_2^\star \approx (0,0,1).$$

Example 10. *A zero-sum 2-player polynomial game in* 4×4 *variables on the simplex*

$$x_i \in \Delta^3 \quad \forall i \in \{1,2\}.$$

The nonconcave-nonconvex utility function [45, Example 6.1 (iii)] is

$$u_1(x_1,x_2) = \sum_{i=1}^{4} \sum_{\substack{j=1 \\ i \neq j}}^{4} (x_{1i}x_{1j} + x_{2i}x_{2j}) - \sum_{i=1}^{4} x_{1i}^2 x_{2i}^2$$

This game has 4 pure NE involving each one-hot vector e_i

$$NE_i:\ x_1^\star \approx (0.25,0.25,0.25,0.25),\ x_2^\star = e_i \quad \forall i \in \{1,2,3,4\}.$$

Example 11. *A zero-sum 2-player polynomial game in* 3×3 *variables on the simplex*

$$x_i \in \Delta^2 \quad \forall i \in \{1,2\}.$$

the nonconcave-nonconvex utility function [45, Example 6.1 (iv)] is

$$u_1(x_1,x_2) = x_{23}^2 x_{11}^2 - x_{11}x_{12}x_{21}x_{22} - x_{11}x_{13}x_{21}x_{23} + x_{21}^2 x_{12}^2 - x_{12}x_{13}x_{22}x_{23} + x_{22}^2 x_{13}^2$$

The utility function has no saddle point.

Example 12. *A zero-sum 2-player game in* 2×2 *variables on the unit hypercube*

$$x_i \in [0,1]^2 \quad \forall i \in \{1,2\}$$

with the concave-nonconvex utility function

$$u_1(x_1,x_2) = -(1 + x_{11} + x_{12} + x_{21} + x_{22})^2 + 4(x_{11} + x_{22} + x_{11}x_{12} + x_{12}x_{21} + x_{21}x_{22})$$

Nie, Yang, and Zhou [45, Example 6.2 (i)] report the pure NE

$$x_1^\star \approx (0.3249, 0.3249),\ x_2^\star = (1,0).$$

Similarly, we recover a pure ϵ*-NE with any feasible* $x_{11} = x_{12}$ *and* $x_2 = (0,1)$.

Example 13. *A zero-sum 2-player game in* 3×3 *variables on the unit hypercube*

$$x_i \in [0,1]^3 \quad \forall i \in \{1,2\}$$

with the nonconcave-nonconvex utility function [45, Example 6.2 (ii)]

$$\begin{aligned} u_1(x_1, x_2) = &- x_{11} - x_{12} - x_{13} - x_{21} - x_{22} - x_{23} - x_{22}^2 x_{11}^2 \\ &- x_{23}^2 x_{11}^2 + x_{21}^2 x_{12}^2 - x_{23}^2 x_{12}^2 + x_{21}^2 x_{13}^2 + x_{22}^2 x_{13}^2 \end{aligned}$$

The utility function has no saddle point.

Example 14. *A zero-sum 2-player game in* 3×3 *variables on the hypercube*

$$x_i \in [-1,1]^3 \quad \forall i \in \{1,2\}$$

with the nonconcave-nonconvex utility function [45, Example 6.3 (i)]

$$u_1(x_1, x_2) = -\sum_{i=1}^{3} (x_{1i} + x_{2i}) + \prod_{i=1}^{3} (x_{1i} - x_{2i})$$

This game has three pure Nash equilibria:

$$\begin{aligned} NE_1 &: x_1^\star = (-1,-1,1), x_2^\star = (1,1,1), \\ NE_2 &: x_1^\star = (-1,1,-1), x_2^\star = (1,1,1), \\ NE_3 &: x_1^\star = (1,-1,-1), x_2^\star = (1,1,1). \end{aligned}$$

Example 15. *A zero-sum 2-player game in* 3×3 *variables on the hypercube*

$$x_i \in [-1,1]^3 \quad \forall i \in \{1,2\}$$

with the nonconcave-nonconvex utility function [45, Example 6.3 (ii)]

$$u_1(x_1, x_2) = -\sum_{i=1}^{3} x_{2i}^2 + \sum_{i=1}^{3} x_{1i}^2 - \sum_{\substack{i=1 \\ i<j}}^{3} \sum_{j=1}^{3} (x_{1i} x_{2j} - x_{1j} x_{2i})$$

This game has the following pure Nash equilibrium:

$$x_1^\star = (-1,1,-1), x_2^\star = (-1,1,-1).$$

Example 16. *A zero-sum 2-player game in* 3×3 *variables on the sphere*

$$x_i \in \left\{ z \in \mathbb{R}^3 \mid \|z\| = 1 \right\} \quad \forall i \in \{1,2\}$$

with the utility function [45, Example 6.4 (i)]

$$\begin{aligned} u_1(x_1, x_2) = &- x_{11}^3 - x_{12}^3 - x_{13}^3 - x_{21}^3 - x_{22}^3 - x_{23}^3 \\ &- 2(x_{11} x_{12} x_{21} x_{22} + x_{11} x_{13} x_{21} x_{23} + x_{12} x_{13} x_{22} x_{23}) \end{aligned}$$

This game has the following 9 pure NE:

$$NE_{ij} : \; x_1^\star = e_i, \, x_2^\star = e_j \quad \forall i, j \in \{1,2,3\}.$$

Example 17. *A zero-sum 2-player game in* 3×3 *variables on the sphere*

$$x_i \in \left\{z \in \mathbb{R}^3 \mid \|z\| = 1\right\} \quad \forall i \in \{1,2\}$$

with the utility function [45, Example 6.4 (ii)]

$$\begin{aligned} u_1(x_1,x_2) = &- x_{11}^2 x_{21}^2 - x_{12}^2 x_{22}^2 - x_{13}^2 x_{23}^2 - x_{11}^2 x_{22} x_{23} - x_{12}^2 x_{21} x_{23} \\ &- x_{13}^2 x_{21} x_{22} - x_{21}^2 x_{12} x_{13} - x_{22}^2 x_{11} x_{13} - x_{23}^2 x_{11} x_{12} \end{aligned}$$

The utility function has no saddle points.

Example 18. *A zero-sum 2-player game in* 3×3 *variables on the ball*

$$x_i \in \left\{z \in \mathbb{R}^3 \mid \|z\| \le 1\right\} \quad \forall i \in \{1,2\}$$

with the nonconcave-convex utility function [45, Example 6.5]

$$u_1(x_1,x_2) = -x_{11}^2 x_{21} - 2x_{12}^2 x_{22} - 3x_{13}^2 x_{23} + x_{11} + x_{12} + x_{13}$$

This game has the following pure Nash equilibrium:

$$NE_3 : x_1^\star \approx (0.7264, 0.4576, 0.3492), x_2^\star \approx (0.6883, 0.5463, 0.4772).$$

Example 19. *A zero-sum 2-player game in* 3×3 *variables on the intersection of the nonnegative orthant and the sphere*

$$x_i \in \left\{z \in \mathbb{R}^3_+ \mid \|z\| = 1\right\} \quad \forall i \in \{1,2\}$$

with the utility function [45, Example 6.6]

$$u_1(x_1,x_2) = -x_{11}^2 x_{22} x_{23} - x_{21}^2 x_{12} x_{13} - x_{12}^2 x_{21} x_{23} - x_{22}^2 x_{11} x_{13} - x_{13}^2 x_{21} x_{22} - x_{23}^2 x_{11} x_{12}$$

The utility function has no saddle points.

Example 20. *A zero-sum 2-player game in* 5×5 *variables on the simplex*

$$x_i \in \Delta^4 \quad \forall i \in \{1,2\}.$$

The utility function is defined by [45, Example 6.10]

$$u_1(x_1,x_2) = x_1^\top A_1 x_1 + x_2^\top A_2 x_2 + x_1^\top B x_2$$

where the matrix parameters A_1, A_2, *and* B *are*

$$A_1 = \begin{bmatrix} -4 & 4 & 0 & 3 & -4 \\ 3 & 4 & 3 & -4 & -5 \\ -3 & 0 & -2 & 0 & 4 \\ -4 & -4 & -1 & 3 & -5 \\ 4 & 1 & -3 & 0 & -5 \end{bmatrix}, A_2 = \begin{bmatrix} -4 & 4 & 1 & 0 & 1 \\ -2 & -4 & 2 & -3 & 1 \\ -3 & 1 & 1 & 4 & 4 \\ 3 & -4 & 0 & 1 & -2 \\ -1 & -3 & -1 & 3 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} -2 & -4 & -2 & -5 & 3 \\ 0 & 0 & 2 & 4 & 2 \\ 0 & -4 & -1 & -5 & 3 \\ 1 & -3 & -4 & 0 & -3 \\ 3 & -1 & -5 & 4 & -4 \end{bmatrix}$$

The game has the following two NE:

$$NE_1 : x_1^\star = (0,1,0,0,0), x_2^\star = (1,0,0,0,0),$$
$$NE_2 : x_1^\star = (0,1,0,0,0), x_2^\star = (0,1,0,0,0).$$

A.3. Miscellaneous Games

The following is a list of example games appearing in papers and books by various authors, which we loosely grouped by the research topic.

A.3.1. Games from Classical Literature

The following games appear in books from the 1950s and 60s, where they were used as demonstrations that the solutions of continuous games may be supported on anything from singleton sets to Cantor sets and whole intervals.

Example 21. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\},$$

with the utility function [16, pg. 110]

$$u_1(x_1, x_2) = (x_{11} - x_{21})^2$$

The Nash equilibrium is $\mu_1^\star = \mathcal{U}\{0,1\}$; $\mu_2^\star = \delta_{0.5}$ *and the value is* $\frac{1}{4}$.

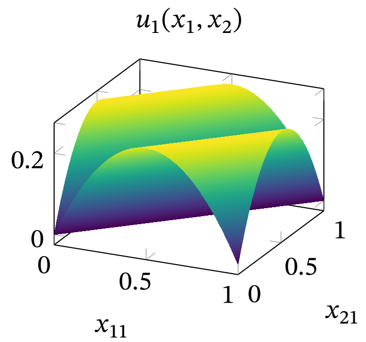
Example 22. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\},$$

with the utility function

$$u_1(x_1, x_2) = (1 - |x_{21} - x_{11}|)\,|x_{21} - x_{11}|$$

Dresher [16, pg. 111] proved that the only equilibrium of this game has an infinite support: $\mu_i^\star = \mathcal{U}[0,1]$, *and its value is* $\frac{1}{6}$.



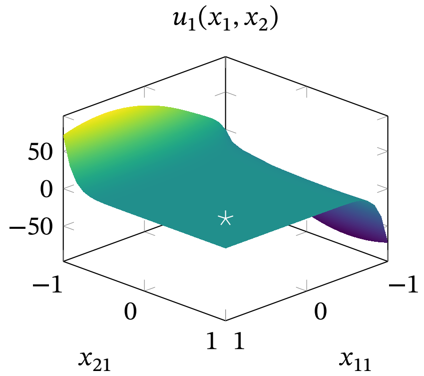


Figure A.1.: 12-degree polynomial game on a hypercube with a prescribed solution at 0.5, 0.5 constructed using the method described by Gale and Gross [20] with parameters $\alpha_i = (1, 0)$, $\sigma_i = (0.5)$, and $\mu_i = (1) \quad \forall i \in \{1, 2\}$.

Example 23. *A two-player zero-sum game in $m_1 \times m_2$ variables on compact subsets of real numbers*

$$x_i \in X_i \subset \mathbb{R}^{m_i} \quad \forall i \in \{1, 2\}.$$

The polynomial utility function is defined by

$$\begin{aligned} u(x_1, x_2) = & f(x_1)\phi(x_1)\left(g(x_2)\phi(x_1) + \sum_{i=1}^{c_2}(g_i(x_2) - \mu_{2i})\phi_i(x_1)\right) \\ & - g(x_2)\psi(x_2)\left(f(x_1)\psi(x_2) + \sum_{i=1}^{c_1}(f_i(x_1) - \mu_{1i})\psi_i(x_2)\right) \\ & - (f(x_1)\phi(x_1))^2 + (g(x_2)\psi(x_2))^2 \end{aligned}$$

where

$$f(x_1) = \prod_{s \in \sigma_1} \|x_1 - s\|^2, \qquad \phi(x_1) = \prod_{a \in \alpha_1} \|x_1 - a\|^2,$$

$$f_i(x_1) = \prod_{\substack{s \in \sigma_1 \\ s \neq \sigma_{1i}}} \frac{\|z - s\|^2}{\|\sigma_{1i} - s\|^2}, \qquad \phi_i(x_1) = \prod_{\substack{a \in \alpha_1 \\ a \neq \alpha_{1i}}} \|x_1 - a\|^2;$$

with g and ψ being defined analogously for x_2, σ_2, μ_2, and α_2.

This game has the property that the parameters σ and μ define a unique mixed Nash equilibrium supported on $\sigma_i \in X_i^{c_i}$ weighted by $\mu_i \in \Delta^{c_i - 1}$, for any set of cluster points $\alpha_i \in X_i^{c_{-i}+1}$ that do not overlap with the supports.

This construction of polynomial games with a unique prescribed solution was discovered by Gale and Gross [20]. The generated games may be challenging to solve numerically given their large degrees and relative flatness near the equilibrium as shown in figure A.1.

Example 24. *A two-player zero-sum game in 1×1 variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\}.$$

The utility function is defined by

$$u_1(x_1, x_2) = \frac{1}{1 + \lambda (x_{11} - x_{21})^2}$$

where $\lambda > 0$. Karlin [31, Example 1] proved that the game has finitely supported optimal strategies.

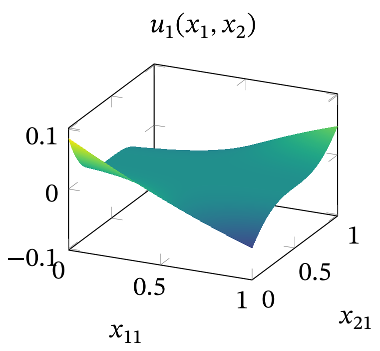
Example 25. *A two-player zero-sum game in 1×1 variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\}.$$

The utility is defined by the rational function

$$u_1(x_1, x_2) = (x_{21} - 0.5)\left(\frac{1 + (x_{21} - 0.5)^2 (x_{11} - 0.5)}{1 + (x_{21} - 0.5)^4 (x_{11} - 0.5)^2}\right) - \frac{x_{21} - 0.5}{1 + (x_{21} - 0.5)^4 \left(\frac{1}{3} x_{11} - 0.5\right)}$$

Karlin [31, Example 2] proved that the unique optimal strategy is the Cantor distribution.



Example 26. *A two-player zero-sum game in 1×1 variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\}.$$

The utility is defined by [31, Problem 1]:

$$u_1(x_1, x_2) = \frac{2(x_{11} + x_{21})}{(1 + 2x_{11})(1 + 2x_{21})}$$

The game has a Nash equilibrium where both players play $\mu^\star(z) = \frac{1+2z}{2}$.

Example 27. *A two-player zero-sum game in 1×1 variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\}.$$

The utility is defined by

$$u_1(x_1, x_2) = \frac{(1 + x_{11})(1 + x_{21})}{(1 + x_{11} x_{21})^2}$$

Karlin [31, Problem 2] showed that the value of this game is $\frac{1}{\ln 2}$ when both players play $\mu^\star(z) = \frac{\ln(1+z)}{\ln 2}$.

A.3.2. Experiments on Games with Interesting Weaker Solution Concepts

The following examples are from papers that study algorithms recovering weaker solution concepts, such as min-max critical points, due to the potential non-existence and the complexity of computing Nash Equilibria. Examples ... were used to demonstrate the failure of convergence of gradient-based algorithms.

Example 28. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\},$$

with the utility function

$$u_1(x_1, x_2) = (-0.5 + x_{11})(-0.5 + x_{21}).$$

This game was studied by Daskalakis et al. [15, Appendix D]. The game has a value of 0 *and a pure Nash equilibrium at* $(x_{11}, x_{21}) = (0.5, 0.5)$*.*

Example 29. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\},$$

with the utility function [15, Appendix E]

$$u_1(x_1, x_2) = \left(-4x_{11}^2 + \frac{x_{21}^4}{10} + \left(-3x_{11} + x_{21} + \frac{x_{11}^3}{20}\right)^2\right) e^{\frac{1}{100}\left(-x_{11}^2 - x_{21}^2\right)}$$

The only local min-max equilibrium is $(x_{11}, x_{21}) = (0,0)$*. This game was also studied by Chinchilla, Yang, and Hespanha [13], Wang, Zhang, and Ba [65], and Zheng et al. [70].*

Example 30. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [0,1] \quad \forall i \in \{1,2\}.$$

The utility function is defined by

$$u_1(x_1, x_2) = -\left(x_{11} - \frac{1}{2}\right)\left(x_{21} - \frac{1}{2}\right) - \frac{1}{3} e^{-\left(x_{11} - \frac{1}{4}\right)^2 - \left(x_{21} - \frac{3}{4}\right)^2}$$

Mertikopoulos et al. [42, Fig. 1] demonstrated divergent behavior of Mirror Descent *on this non-monotone saddle-point problem. The game has an* ϵ*-NE:*

$$\mu_1^\star \approx \delta_{0.38},$$
$$\mu_2^\star \approx 0.433\delta_0 + 0.567\delta_1$$

achieving a value of 0.247*.*

Example 31. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_{i1} \in [-1, 1] \quad \forall i \in \{1, 2\}.$$

The utility function is

$$u_1(x_1, x_2) = \left(1 - x_{21}^2 + x_{21}^4 x_{11}^2\right)\left(-1 - x_{11}^2 - x_{21}^2 x_{11}^4\right)$$

This non-monotone saddle-point problem was studied by Mertikopoulos et al. [42, (2.2)]. The unique saddle point is $x^\star = (0, 0)$.

Example 32. *A two-player zero-sum game in* 1×1 *variables on the intervals*

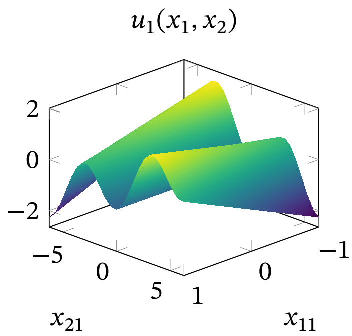
$$x_1 \in [-1, 1]$$
$$x_2 \in [-2\pi, 2\pi]$$

with the utility function [53, Example 1]

$$u_1(x_1, x_2) = -\cos(x_{21}) + 0.2 x_{11} x_{21}$$

The game has the ϵ*-NE:*

$$\mu_1^\star = 0.5\delta_{-1} + 0.5\delta_1,$$
$$\mu_2^\star = 0.5\delta_{-2\pi} + 0.5\delta_{2\pi}.$$



Example 33. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}$$

with the utility function [53, Example 3]

$$u_1(x_1, x_2) = 2x_{11}x_{21} - x_{21}^2 + x_{11}^3$$

The game has the mixed ϵ*-NE:*

$$\mu_1^\star \approx \frac{1}{3}\delta_1 + \frac{2}{3}\delta_{-0.5},$$
$$\mu_2^\star \approx \frac{5}{16}\delta_1 + \frac{11}{16}\delta_{-1}.$$

Example 34. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}$$

with utility function given by

$$u_1(x_1, x_2) = -2x_{11}^2 - 4x_{11}x_{21} + x_{21}^2 - \frac{4}{3}x_{21}^3 + \frac{1}{4}x_{21}^4$$

This example was studied by Zheng et al. [70, Convex-Nonconcave Example] and also Chinchilla, Yang, and Hespanha [13] and Adolphs et al. [3].

Example 35. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}$$

with utility function [70, KŁ-Nonconcave Example] given by

$$u_1(x_1, x_2) = -x_{11}^2 + 4x_{21}^2 + 10\sin^2(x_{21}) - 3\sin^2(x_{21})\sin^2(x_{11})$$

This game has the pure NE $x^\star = (0, 0)$.

Example 36. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-4, 4] \quad \forall i \in \{1, 2\}$$

with utility function [70, Bilinearly-coupled Minimax] given by

$$u_1(x_1, x_2) = -f(x_{11}) - Ax_{11}x_{21} + f(x_{21})$$

where $f(z) = (z + 1)(z - 1)(z + 3)(z - 3)$. *This example was also studied by Grimmer et al. [25].*

Example 37. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1.5, 1.5] \quad \forall i \in \{1, 2\}$$

with utility function [70, "Forsaken" example] given by

$$u1(x, y) = -x_{11}(x_{21} - 0.45) - \phi(x_{11}) + \phi(x_{21})$$

where $\phi(z) = \frac{1}{4}z^2 - \frac{1}{2}z^4 + \frac{1}{6}z^6$. *This problem was also studied by Hsieh, Mertikopoulos, and Cevher [29, Example 5.2]. The game has the mixed* ϵ*-NE:*

$$\mu_1^\star = 0.5\delta_{-1.31} + 0.5\delta_{1.31},$$
$$\mu_2^\star = 0.328\delta_{-1.31} + 0.672\delta_{1.31}.$$

A.3.3. Experiments on Games by Parillo, Stein and Ozdaglar

The following examples are games where the strategy spaces for each player are intervals. The games were originally solved using Lasserre hierarchies, discretization techniques, or were used to demonstrate methods for obtaining bounds on the size of support of equilibria.

Example 38. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}.$$

The utility function is [50, Example 2.1]

$$u_1(x_1, x_2) = (x_{11} - x_{21})^2$$

The Nash equilibrium is $\mu_1^\star = \mathcal{U}\{-1, 1\}$; $\mu_2^\star = \delta_0$.

Example 39. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}.$$

The utility function is [50, Example 3.1]

$$u_1(x_1, x_2) = -x_{21} - x_{11}^2 + 2x_{21}^2 x_{11}$$

The game has the following pure NE and a value of $-\frac{3\sqrt[3]{2}}{8}$*:*

$$\begin{aligned} x_1^\star &= 4^{-\frac{2}{4}}, \\ x_2^\star &= 4^{-\frac{1}{4}}. \end{aligned}$$

Example 40. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}.$$

The utility function is [50, Example 3.2]

$$u_1(x_1, x_2) = -x_{21} - 2x_{11}^2 + 5x_{11}x_{21} - 2x_{21}^2 x_{11}$$

The game has the following mixed NE with a value of 0.48*:*

$$\begin{aligned} \mu_1^\star &\approx \delta_{0.2}, \\ \mu_2^\star &\approx 0.78\delta_1 + 0.22\delta_{-1}. \end{aligned}$$

Example 41. *A general-sum 2-player game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}$$

with utility functions [60, Example 3.1] given by

$$\begin{aligned} u_1(x_1, x_2) &= 0.554 + 1.36x_{11} - 1.2x_{21} + 0.596x_{11}^2 + 2.072x_{11}x_{21} - 0.394x_{21}^2 \\ u_2(x_1, x_2) &= 1.918x_{11}x_{21} - 1.886 - 1.232x_{11} + 0.842x_{21} - 0.108x_{11}^2 - 1.044x_{21}^2. \end{aligned}$$

The game has an equilibrium $x^\star = (1, 1)$.

Example 42. *A general-sum 2-player game in* 1×1 *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2\}$$

with utility functions [59, Example 2.3] given by

$$\begin{aligned} u_1(x_1, x_2) &= -x_{11} + 2x_{11}x_{21} - 2x_{11}^3 + 3x_{21}^3 - 3x_{21}^2 x_{11}^2 \\ u_2(x_1, x_2) &= 4x_{21} - x_{11}^2 + x_{11}^2 x_{21} - 4x_{21}^3 + 2x_{21}^2 x_{11}^2 \end{aligned}$$

The game has the mixed NE:

$$\begin{aligned} \mu_1^\star &\approx 0.5532\delta_{-1} + 0.4468\delta_{0.1149}, \\ \mu_2^\star &\approx \delta_{0.7166}. \end{aligned}$$

Example 43. *A general-sum 2-player game in* 1×1 *variables on the interval*

$$x_i \in [-\pi, \pi] \quad \forall i \in \{1, 2\}$$

with utility functions [59, Example 2.4] given by

$$\begin{aligned} u_1(x_1, x_2) &= \cos(x_{11} - x_{21}) \\ u_2(x_1, x_2) &= \cos(-0.5 + x_{11} - x_{21}) \end{aligned}$$

The game has mixed equilibria where each player plays θ_i *and* $\theta_i + \pi$ *with equal probability, where* θ_i *is arbitrary.*

Example 44. *A 3-player game in* $1 \times 1 \times 1$ *variables on the interval*

$$x_i \in [-1, 1] \quad \forall i \in \{1, 2, 3\}$$

with utility functions [59, Example 3.10] given by

$$\begin{aligned} u_1(x_1, x_2, x_3) &= 1 + 2x_{11} + 3x_{11}^2 + 2x_{21}x_{31} + 4x_{11}x_{21}x_{31} + 6x_{11}^2x_{21}x_{31} \\ &\quad + 3x_{31}^2x_{21}^2 + 6x_{31}^2x_{21}^2x_{11} + 9x_{31}^2x_{21}^2x_{11}^2 \\ u_2(x_1, x_2, x_3) &= 7 + 2x_{11} + 2x_{21} + 3x_{11}^2 + 4x_{11}x_{21} + 3x_{31}^2 + 6x_{11}^2x_{21} \\ &\quad + 6x_{31}^2x_{11} + 9x_{31}^2x_{11}^2 \\ u_3(x_1, x_2, x_3) &= -x_{31} - 2x_{11}x_{31} - 2x_{21}x_{31} - 3x_{11}^2x_{31} - 4x_{11}x_{21}x_{31} \\ &\quad - 3x_{31}^2x_{21} - 6x_{11}^2x_{21}x_{31} - 6x_{31}^2x_{11}x_{21} - 9x_{31}^2x_{11}^2x_{21} \end{aligned}$$

The game has the pure NE $x^\star = (1, 1, -0.5)$*.*

A.3.4. Experiments on Applied Games and Others

The following games are a mix of game theory applied to communication networks and security games, as well as others that did not fit elsewhere.

Example 45. *A general-sum 2-player game in* 1×1 *variables constrained to the torus*

$$x_{i1} \in [-\pi, \pi] \quad \forall i \in \{1, 2\}$$

with utilities parameterized by $\phi = \left(0, \frac{\pi}{8}\right), \alpha = (1, 1.5)$*:*

$$\begin{aligned} u_1(x_1, x_2) &= \alpha_1 \cos(x_{11} - \phi_1) - \cos(x_{11} - x_{21}) \\ u_2(x_1, x_2) &= \alpha_2 \cos(x_{21} - \phi_2) - \cos(x_{21} - x_{11}) \end{aligned}$$

This game appeared in a paper by Chasnov et al. [12, p. 5.2], who reported pure equilibria at $(-1.063, 1.014)$ *and* $(1.408, -0.325)$*.*

Example 46. *A two-player zero-sum game in* 1×1 *variables constrained to the interval*

$$x_{i1} \in [0, 1] \quad \forall i \in \{1, 2\}.$$

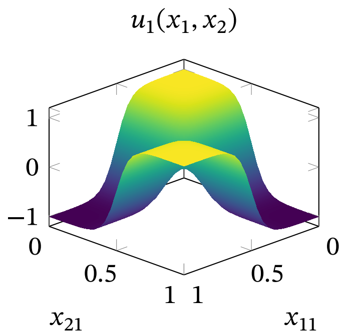
The utility function is

$$u_1(x_1, x_2) = \sigma(x_2)^\top A \sigma(x_1)$$

where $\sigma(x)$ *is the* softmax *vector*

$$\sigma(x) = \left[\frac{e^{10x}}{e^{10x} + e^{10(1-x)}}, \frac{e^{10(1-x)}}{e^{10x} + e^{10(1-x)}} \right]$$

This is a continuous version of the classical matching-pennies game, and it was studied by Chasnov et al. [12, B.1], who reported a pure equilibrium at $(x_{11}, x_{21}) = (0.5, 0.5)$.



Example 47. *A general-sum 2-player game in* 1×1 *variables on the interval*

$$x_i \in [-\pi, \pi] \quad \forall i \in \{1, 2\}.$$

The utility functions are given by

$$\begin{aligned} u_1(x_1, x_2) &= \cos(x_{11}) - \alpha_1 \cos(x_{11} - x_{21}) \\ u_2(x_1, x_2) &= \cos(x_{21}) - \alpha_2 \cos(-x_{11} + x_{21}). \end{aligned}$$

Ratliff, Burden, and Sastry [52, Location Game] studied this game with $\alpha = [1, 1.05]$, *in which case the game has two NE: at* (x_1, x_2) *equal to* $(1, -1.1)$, *or* $(-1, 1.1)$.

Example 48. *A 3-player polynomial network game in* $1 \times 1 \times 1$ *variables on the interval*

$$x_{i1} \in [0, 1] \quad \forall i \in \{1, 2, 3\}.$$

The utilities [33, Example 5] are

$$\begin{aligned} u_1(x_1, x_2, x_3) &= -x_{21} - 2x_{31} - 2x_{11}^2 + 5x_{11}x_{21} - 4x_{11}x_{31} - 2x_{21}^2 x_{11} \\ u_2(x_1, x_2, x_3) &= x_{21} - 2x_{11}^2 - 5x_{11}x_{21} - 2x_{21}^2 + 5x_{21}x_{31} + 2x_{21}^2 x_{11} - 2x_{31}^2 x_{21} \\ u_3(x_1, x_2, x_3) &= 2x_{31} + 4x_{11}^2 + 4x_{11}x_{31} + 2x_{21}^2 - 5x_{21}x_{31} + 2x_{31}^2 x_{21} \end{aligned}$$

The game has the following ϵ*-NE:*

$$\begin{aligned} \mu_1^\star &= 0.291\delta_{-0.056} + 0.191\delta_{-0.066} + 0.518\delta_{-0.06} \\ \mu_2^\star &= 0.232\delta_{0.347} + 0.147\delta_{0.359} + 0.622\delta_{0.352} \\ \mu_3^\star &= 0.281\delta_{-1.0} + 0.719\delta_{1.0}. \end{aligned}$$

Example 49. *A 4-player game in* $4 \times 4 \times 2 \times 2$ *variables constrained to the scaled simplexes:*

$$x_i \in \left\{ z \in \mathbb{R}^4 \;\middle|\; z_i \geq 0,\ \sum_{i=1}^{m} z_i = a_i \right\} \quad \forall i \in \{1, 2\},$$
$$x_i \in \left\{ z \in \mathbb{R}^2 \;\middle|\; z_i \geq 0,\ \sum_{i=1}^{m} z_i = 1 \right\} \quad \forall i \in \{3, 4\}.$$

The utilities are defined by

$$\begin{aligned}
u_1(x_1, x_2, x_3, x_4) &= a_1 - f_1(x_{11}, x_{31}) - f_2(x_{12}, x_{32}) - f_3(x_{13}, x_{41}) - f_4(x_{14}, x_{42}) - o \\
u_2(x_1, x_2, x_3, x_4) &= a_2 - f_1(x_{21}, x_{31}) - f_2(x_{22}, x_{32}) - f_3(x_{23}, x_{41}) - f_4(x_{24}, x_{42}) - o \\
u_3(x_1, x_2, x_3, x_4) &= f_1(x_{11}, x_{31}) + f_2(x_{12}, x_{32}) + f_1(x_{21}, x_{31}) + f_2(x_{22}, x_{32}) - o \\
u_4(x_1, x_2, x_3, x_4) &= f_3(x_{13}, x_{41}) + f_4(x_{14}, x_{42}) + f_3(x_{23}, x_{41}) + f_4(x_{24}, x_{42}) - o
\end{aligned}$$

where $a = [0.5, 0.8]$, $o = 0.325$, *and* f *are the* efficiency *functions:*

$$\begin{aligned}
f_1(u, v) &= (1 - v^2)0.7u^2 \\
f_2(u, v) &= (1 - v)^2(1.4u - 0.7u^2) \\
f_3(u, v) &= (1 - v^2)(1.8u - 0.9u^2) \\
f_4(u, v) &= (1 - v)^2 0.9u^2
\end{aligned}$$

This polynomial security game was studied by Kroupa, Vannucci, and Votroubek [33, Example 6].

Example 50. *A general-sum 2-player game in* $(m + 1) \times 1$ *variables on the unit hypercubes*

$$\begin{aligned}
x_1 &\in [0, 1]^{m+1} \\
x_2 &\in [0, 1].
\end{aligned}$$

The utility functions are given by

$$\begin{aligned}
u_1(x_1, x_2) &= \sum_{\substack{i=0,\dots,m \\ j=i+1}} x_{1j} \binom{m}{i} x_{21}^i (1 - x_{21})^{m-i} - \frac{\gamma}{2^m} \sum_{\substack{i=0,\dots,m \\ j=i+1}} x_{1j} \binom{m}{i} - 1 \\
u_2(x_1, x_2) &= -(x_{21} - 0.8)^2 + \sum_{\substack{i=0,\dots,m \\ j=i+1}} \binom{m}{i} (1 - x_{1j}) x_{21}^i (1 - x_{21})^{m-i}.
\end{aligned}$$

This example is based on the hypothesis testing game *by Yasodharan and Loiseau [69].*

Example 51. *A 6-player game in* $1 \times 1 \times 1 \times 1 \times 1 \times 1$ *variables on the interval*

$$x_{i1} \in [x_{\min}, x_{\max}] \quad \forall i \in \{1, \dots, 6\}.$$

The utility functions are defined by:

$$u_i(x) = -\eta_i x_{i1} + \beta_i \left(\ln \left(1 + a_i \frac{\gamma_i(x)}{1 - \Phi_{ii}\gamma_i(x)} \right) - x_{i1} \right) \quad \forall i \in \{1, \dots, 6\},$$

where η_i, β_i, and a_i are channel-specific parameters. The parameter $\gamma_i(x)$ represents the OSNR [47], defined as

$$\gamma_i(x) = \frac{x_{i1}}{n_0 + \sum_j \Phi_{ij} x_{j1}},$$

for a given system matrix Φ and input noise n_0. The numerical setting is adopted from an article by Nguyen et al. [44]. Specifically, $n_0 = 0.43 \times 10^{-6}$, $\eta_1 = \cdots = \eta_6 = 1$,

$$\beta = [0.5, 0.51, 0.52, 0.3, 0.31, 0.32],$$
$$a = [0.261, 0.494, 0.107, 0.366, 0.208, 0.305],$$

$x_{\min} = 0.2$, $x_{\max} = 2$, and the matrix

$$\Phi = \begin{bmatrix} 7.463 & 7.378 & 7.293 & 7.210 & 7.127 & 6.965 \\ 7.451 & 7.365 & 7.281 & 7.198 & 7.115 & 6.953 \\ 7.438 & 7.353 & 7.269 & 7.186 & 7.103 & 6.942 \\ 7.427 & 7.342 & 7.258 & 7.175 & 7.093 & 6.931 \\ 7.409 & 7.324 & 7.240 & 7.157 & 7.075 & 6.914 \\ 7.387 & 7.303 & 7.219 & 7.136 & 7.055 & 6.894 \end{bmatrix} \times 10^{-5}.$$

The game has a Nash equilibrium $x^\star \approx [0.3329, 0.3375, 0.3412, 0.2305, 0.2361, 0.2421]$.

A.3.5. Rational Saddle Point Games

The following examples are the saddle point problems of rational polynomials studied by Zhou, Wang, and Zhao [71], which we interpret here as two-player zero-sum games where each player controls multiple variables.

Example 52. *A two-player zero-sum game in 2×2 variables on the unit hypercube*

$$x_i \in [0,1]^2 \quad \forall i \in \{1,2\}.$$

The rational utility function [71, Example 5.3 (i)] is given by

$$u_1(x_1, x_2) = \frac{-x_{11}^2 - x_{12}^2 + x_{21}^2 + x_{22}^2}{x_{11} + x_{22} + 1}$$

This game has a pure Nash equilibrium at

$$x_1^\star = (0,0),\ x_2^\star = (0,0).$$

Example 53. *A two-player zero-sum game in 3×3 variables on the unit hypercube*

$$x_i \in [0,1]^3 \quad \forall i \in \{1,2\}.$$

The rational utility function [71, Example 5.3 (ii)] is given by

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^3 (x_{1i} + x_{2i}) - \sum_{\substack{i=1 \\ i<j}}^3 \sum_{j=1}^3 (x_{1i}^2 x_{2j}^2 - x_{2i}^2 x_{1j}^2)}{x_{11}^2 + x_{22}^2 + x_{13}x_{23} + 1}$$

This game has no pure Nash equilibria.

Example 54. *A zero-sum 2-player game in* 3×3 *variables on the nonnegative orthant*

$$x_i \in \mathbb{R}^3_+ \quad \forall i \in \{1, 2\}$$

with the utility function [71, Example 5.4 (i)]

$$u_1(x_1, x_2) = \frac{-x_{11}^4 - x_{12}^4 - x_{13}^4 + x_{21}^4 + x_{22}^4 + x_{23}^4}{x_{11} + x_{21} + 1}$$

The game has a pure Nash equilibrium:

$$x_1^\star \approx (0.0015, 0.0015, 0.0015),$$
$$x_2^\star \approx (0.0015, 0.0015, 0.0015).$$

A.4. Optimization Test Functions

Following the idea of Adam et al. [2], who used a constrained-optimization example as the utility function of a two-player zero-sum game in Example 55, we constructed games based on the dataset of optimization test functions by Surjanovic and Bingham [62]. Specifically, we selected one example from each shape category: many local minima, bowl-shaped, plate-shaped, valley-shaped, steep ridges, and other.

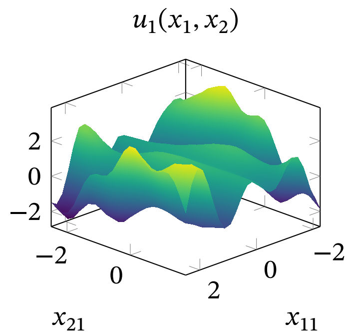
Example 55. *A two-player zero-sum game in* 1×1 *variables constrained to the intervals*

$$x_{11} \in [-2.25, 2.5]$$
$$x_{21} \in [-2.5, 1.75]$$

with the utility given by the Townsend *function*

$$u_1(x_1, x_2) = x_{11} \sin(3x_{11} + x_{21}) + \cos^2((x_{11} - 0.1)\, x_{21})$$

This game was studied by Adam et al. [2, Townsend].



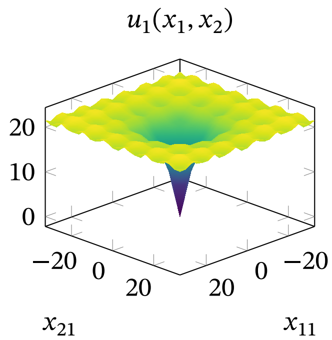
Example 56. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-32.768, 32.768] \quad \forall i \in \{1, 2\}$$

with the utility function [62, Ackley]

$$u_1(x_1, x_2) = -ae^{-b\sqrt{\frac{1}{2}\sum_{i=1}^2 x_i^2}} - e^{\frac{1}{2}\sum_{i=1}^2 \cos(cx_i)} + a + e$$

where $a = 20$, $b = 0.2$, *and* $c = 2\pi$.

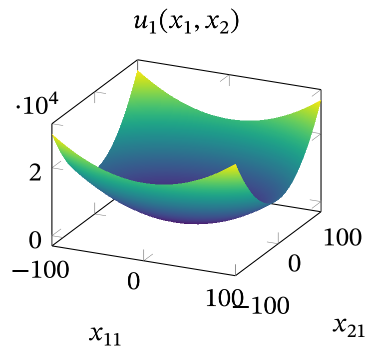


Example 57. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-100, 100] \quad \forall i \in \{1, 2\}$$

with the utility function [62, Bohachevsky 1]

$$u_1(x_1, x_2) = -0.3\cos(3\pi x_1) - 0.4\cos(4\pi x_2) + x_1^2 + 2x_2^2 + 0.7$$



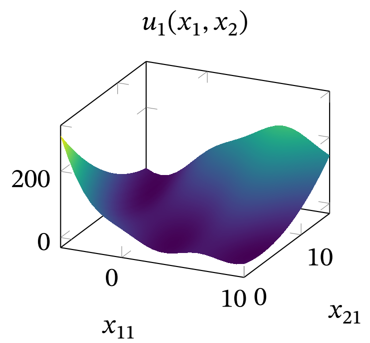
Example 58. *A two-player zero-sum game in* 1×1 *variables on the intervals*

$$x_1 \in [-5, 10]$$
$$x_2 \in [0, 15]$$

with the utility function [62, Branin]

$$u_1(x_1, x_2) = a(x_2 - bx_1^2 + cx_1 - r)^2 + s(1-t)\cos(x_1) + s$$

where $a = 1$, $b = \frac{5.1}{4\pi^2}$, $c = \frac{5}{\pi}$, $r = 6$, $s = 10$, $t = \frac{1}{8\pi}$.

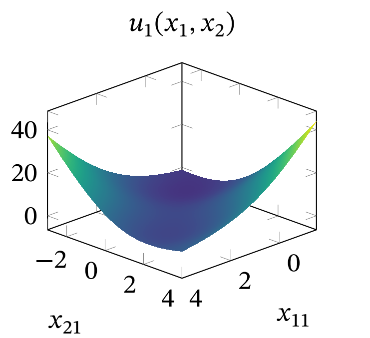


Example 59. *A two-player zero-sum game in* 1×1 *variables on the intervals*

$$x_1 \in [-1.5, 4]$$
$$x_2 \in [-3, 4]$$

with the utility function [62, McCormick]

$$u_1(x_1, x_2) = \sin(x_1 + x_2) + (x_1 - x_2)^2 - \frac{3x_1 + 5x_2 + 2}{2}$$



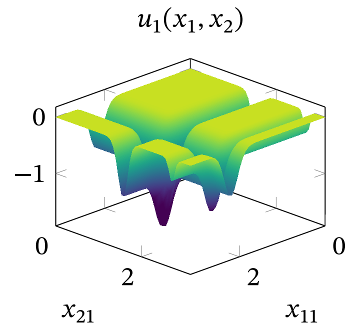
Example 60. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [0, \pi] \quad \forall i \in \{1, 2\}$$

with the utility function [62, Michalewicz]

$$u_1(x_1, x_2) = -\sum_{i=1}^{2} \sin(x_i) \sin^{2m}\left(\frac{ix_i^2}{\pi}\right)$$

where $m = 20$.

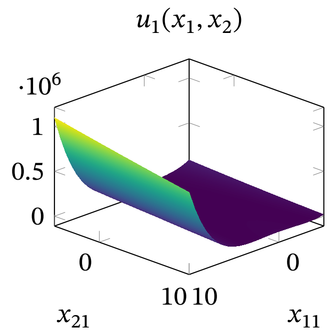


Example 61. *A two-player zero-sum game in* 1×1 *variables on the interval*

$$x_i \in [-5, 10] \quad \forall i \in \{1, 2\}$$

with the utility function [62, Rosenbrock]

$$u_1(x_1, x_2) = 100(y - x^2)^2 + (x - 1)^2$$



A.5. Discontinuous Games with Known Equilibria

For completeness, we finish with a small number of games that either contain discontinuities or have non-compact strategy sets. Neither equilibria nor best responses necessarily exist in such games; for this reason, we include only examples with known solutions.

Example 62. *A two-player zero-sum game in* $m \times m$ *variables on the simplex*

$$x_i \in \Delta^{m-1} = \left\{ z \in \mathbb{R}^m \;\middle|\; z_j \geq 0, \; \sum_{j=1}^{m} z_j = 1 \right\} \quad \forall i \in \{1, 2\}$$

The utility is defined by the discontinuous function

$$u_1(x_1, x_2) = \sum_{i=1}^{m} \operatorname{sign}\left(x_{1j} - x_{2j}\right)$$

This is an infinite version of the classical Colonel Blotto *game [36, 34, 26, 24], in which all isolated fronts have equal value. Pure strategies exist only for* $m \in \{1, 2\}$, *in which case every strategy is a tie; or for* $m = 3$, *when players should allocate* $\frac{1}{3}$ *to every front. Otherwise, Roberson [55] proved that players must play a strategy with marginals* $\mu_{ij}^{\star} = \mathcal{U}\left[0, \frac{2}{m}\right]$ *for every front* j.

is that right?

A.5.1. Polynomial Saddle Point Games

The following polynomial games are played over unbounded strategy sets.

Example 63. *A zero-sum 2-player game in* 4×4 *variables on the nonnegative orthant*

$$x_i \in \mathbb{R}^4_+ \quad \forall i \in \{1,2\}$$

with the utility function [45, Example 6.7]

$$\begin{aligned} u_1(x_1,x_2) = &- x_{21}(x_{12}+x_{13}+x_{14}-1)^2 - x_{22}(x_{11}+x_{13}+x_{14}-2)^2 \\ &- x_{23}(x_{11}+x_{12}+x_{14}-3)^2 + x_{24}(x_{11}+x_{12}+x_{13}-4)^2 \\ &+ x_{11}(x_{22}+x_{23}+x_{24}-1)^2 + x_{12}(x_{21}+x_{23}+x_{24}-2)^2 \\ &- x_{13}(x_{21}+x_{22}+x_{24}-3)^2 + x_{14}(x_{21}+x_{22}+x_{23}-4)^2 \end{aligned}$$

The game has a pure Nash equilibrium:

$$\begin{aligned} x_1^\star &\approx (1.5075, 0.5337, 0.0000, 0.5018), \\ x_2^\star &\approx (2.4143, 1.1463, 0.0000, 0.0000). \end{aligned}$$

Example 64. *A zero-sum 2-player game in* 3×3 *variables with no constraints*

$$x_i \in \mathbb{R}^3 \quad \forall i \in \{1,2\}.$$

The utility function [45, Example 6.8] is

$$u_1(x_1,x_2) = -\sum_{i=1}^{3}(x_{1i}^4 - x_{2i}^4 + x_{1i} + x_{2i}) - \sum_{i=1}^{3}\sum_{\substack{j=1\\ i\neq j}}^{3} x_{1i}^3 x_{2j}^3$$

The game has a pure Nash equilibrium:

$$\begin{aligned} x_1^\star &\approx -(0.6981, 0.6981, 0.6981), \\ x_2^\star &\approx (0.4979, 0.4979, 0.4979). \end{aligned}$$

Example 65. *A zero-sum 2-player game in* 3×3 *variables subject to the constraints*

$$x_i \in \left\{z \in \mathbb{R}^3 \,\middle|\, z_1 \geq 0, z_1 z_2 \geq 1, z_2 z_3 \geq 1\right\} \quad \forall i \in \{1,2\}$$

which define a non-compact set. The utility function [45, Example 6.9] is

$$u_1(x_1,x_2) = -x_{11}^3 x_{21} - x_{12}^3 x_{22} - x_{13}^3 x_{23} + 3x_{11}x_{12}x_{13} + x_{21}^2 + 2x_{22}^2 + 3x_{23}^2$$

The game has a pure Nash equilibrium:

$$\begin{aligned} x_1^\star &\approx (1.2599, 1.2181, 1.3032), \\ x_2^\star &\approx (1.0000, 1.1067, 0.9036). \end{aligned}$$

A.5.2. Rational Saddle Point Games

The following games were created from the saddle point problems of rational polynomials studied by Zhou, Wang, and Zhao [71]. The utility functions are not continuous on the whole strategy set as the divisor often vanishes on the boundary.

Example 66. *A two-player zero-sum game in* 3×3 *variables on the simplex*

$$x_i \in \Delta^2 \quad \forall i \in \{1, 2\}.$$

The rational utility function [71, Example 5.1] is given by

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}\sum_{j=1}^{3} x_{1i}^2 x_{2j}^2 + \sum_{\substack{i=1\\ i\neq j}}^{3}\sum_{j=1}^{3}(x_{1i}x_{1j} + x_{2i}x_{2j})}{x_{11} + x_{12} + x_{22} + x_{23}}$$

This game has two pure Nash equilibria:

$$NE_1 : x_1^\star \approx (0.3165, 0.3165, 0.3670),\ x_2^\star = (0, 1, 0),$$
$$NE_2 : x_1^\star \approx (0.3165, 0.3165, 0.3670),\ x_2^\star = (0, 0, 1).$$

Example 67. *A two-player zero-sum game in* 3×3 *variables on the hypercube*

$$x_i \in [-1, 1]^3 \quad \forall i \in \{1, 2\}.$$

The rational utility function [71, Example 5.2 (ii)] is given by

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3} x_{1i}^2 + \sum_{i=1}^{3} x_{2i}^2}{\sum_{i=1}^{3} x_{2i}^2 - \sum_{i=1}^{3} x_{1i}^2 + 3}$$

This game has a pure Nash equilibrium at

$$x_1^\star = (0, 0, 0),\ x_2^\star = (0, 0, 0).$$

Example 68. *A two-player zero-sum game in* 2×2 *variables on the ball*

$$x_i \in \left\{z \in \mathbb{R}^2 \mid \|z\| \leq 1\right\} \quad \forall i \in \{1, 2\}$$

The rational utility function [71, Example 5.5 (i)] is

$$u_1(x_1, x_2) = \frac{-x_{11}^2 - x_{12}^2 + 3x_{11}x_{12} + x_{21}^2 + x_{22}^2 - x_{21}x_{22}}{x_{11}x_{21} + 1}$$

This game has no pure Nash equilibria.

Example 69. *A two-player zero-sum game in* 3×3 *variables on the ball*

$$x_i \in \left\{z \in \mathbb{R}^3 \mid \|z\| \leq 1\right\} \quad \forall i \in \{1, 2\}$$

The rational utility function [71, Example 5.5 (ii)] is

$$u_1(x_1, x_2) = \frac{-x_{11}^2 x_{21} - 2x_{12}^2 x_{22} - 3x_{13}^2 x_{23} + x_{11} + x_{12} + x_{13}}{x_{11}x_{21} + 1}$$

This game has a pure Nash equilibrium at

$$x_1^\star \approx (0.4730, 0.4476, 0.3416),\ x_2^\star \approx (0.6708, 0.5585, 0.4879).$$

Example 70. *A two-player zero-sum game in* 3×3 *variables on the sphere*

$$x_i \in \left\{z \in \mathbb{R}^3 \mid \|z\| = 1\right\} \quad \forall i \in \{1, 2\}$$

The rational utility function [71, Example 5.6 (i)] is

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}(x_{1i}^3 + x_{2i}^3) - 2(x_{11}x_{12}x_{21}x_{22} + x_{11}x_{13}x_{21}x_{23} + x_{12}x_{13}x_{22}x_{23})}{x_{11} - x_{21} + 1}$$

The game has no pure NE.

Example 71. *A two-player zero-sum game in* 3×3 *variables on the sphere*

$$x_i \in \left\{z \in \mathbb{R}^3 \mid \|z\| = 1\right\} \quad \forall i \in \{1, 2\}$$

The rational utility function [71, Example 5.6 (ii)] is

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}(x_{1i}^2 + x_{2i}^2 - x_{1i} - x_{2i})}{x_{13} + x_{22} + 2}$$

This game has a pure Nash equilibrium at

$$x_1^\star \approx (0.4082, 0.4082, 0.8165),\ x_2^\star \approx (-0.4082, -0.8165, -0.4082).$$

Example 72. *A two-player zero-sum game in* 3×2 *variables on the intersection of the nonnegative orthant and the n-sphere*

$$x_1 \in \left\{z \in \mathbb{R}_+^3 \mid \|z\| = 1\right\}$$
$$x_2 \in \left\{z \in \mathbb{R}_+^2 \mid \|z\| = 1\right\}$$

The rational utility function is

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}(x_{1i}^2 - x_{1i}) - \sum_{i=1}^{2}(x_{2i} - x_{2i}^2)}{x_{11}^2 - x_{21}^2 + 1}$$

The utility has a discontinuity at the boundary of x_1 *when* $x_{21} = 1$*. Zhou, Wang, and Zhao [71, Example 5.7 (i)] report that this game has no saddle point, although we found the following pure* ϵ*-NE (with* $\epsilon < 10^{-8}$*):*

$$x_1^\star \approx (0.4351314094, 0.6366555805, 0.6366555805),\ x_2^\star \approx (0.5255286571, 0.8507758992).$$

We also show this game reprojected in polar coordinates in Example 73, where we recover the same solution.

Example 73. *A two-player zero-sum game in* 3×2 *variables on the intersection of the nonnegative orthant and the n-sphere in polar coordinates*

$$x_1 \in \left[0, \frac{\pi}{2}\right]^2$$
$$x_1 \in \left[0, \frac{\pi}{2}\right]$$

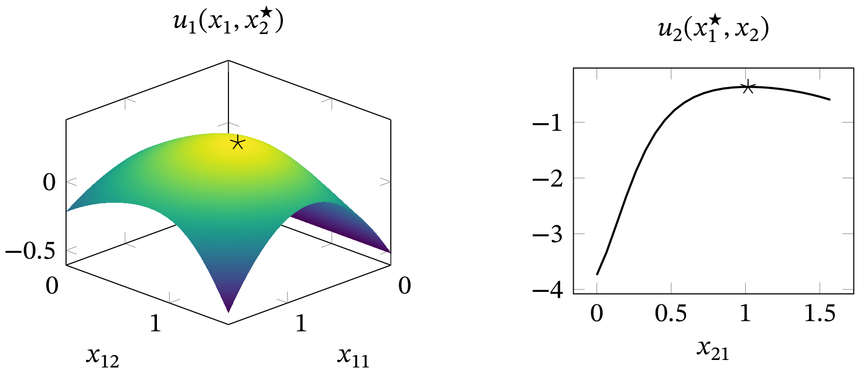


Figure A.2.: The best-response curves of both players at the ϵ-NE of Example 73.

The utility function is

$$u_1(x_1, x_2) = v\left([\sin(x_{11})\cos(x_{12}), \sin(x_{11})\sin(x_{12}), \cos(x_{11})], [\cos(x_{21}), \sin(x_{21})]\right)$$

where

$$v(z_1, z_2) = \frac{-\sum_{i=1}^{3}(z_{1i}^2 - z_{1i}) - \sum_{i=1}^{2}(z_{2i} - z_{2i}^2)}{z_{11}^2 - z_{21}^2 + 1}$$

This example is the polar coordinate reformulation of the example by Zhou, Wang, and Zhao [71, Example 5.7 (i)]. Despite containing a discontinuity at the boundary of x_1 when $x_{21} = 0$ the game has the pure ϵ-NE shown in Fig. A.2:

$$x_1^\star \approx (0.8813, 0.9716),$$
$$x_2^\star \approx 1.0175.$$

Example 74. *A two-player zero-sum game in 3×3 variables on the intersection of the nonnegative orthant and the sphere*

$$x_i \in \left\{z \in \mathbb{R}_+^3 \mid \|z\| = 1\right\} \quad \forall i \in \{1, 2\}$$

with the utility function [71, Example 5.7 (ii)]

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}(x_{1i} - 1)^2 - \sum_{i=1}^{3}(x_{2i} - 1)^2}{x_{11}^2 - x_{21}^2 + 1}$$

The game has the pure NE

$$x_1^\star \approx (0.9519, 0.2167, 0.2167),\ x_2^\star \approx (1, 0, 0).$$

Example 75. *A general-sum 2-player game in 3×3 variables with no constraints*

$$x_i \in \mathbb{R}^3 \quad \forall i \in \{1, 2\}.$$

with the utility function [71, Example 5.8 (i)]

$$u_1(x_1, x_2) = \frac{-x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{21}^2 - x_{22}^2 + x_{11} + x_{12}}{x_{11}^2 + x_{23}^2 + 1}$$

The game has a pure Nash equilibrium:

$$x_1^\star \approx (0.3508, 0.5000, 0.0000),$$
$$x_2^\star \approx (0.0000, 0.0000, 0.0000).$$

Example 76. *A zero-sum 2-player game in* 3×3 *variables subject to the constraints*

$$x_i \in \left\{z \in \mathbb{R}^3 \,\middle|\, z_1 \geq 0, z_1 z_2 \geq 1, z_2 z_3 \geq 1\right\} \quad \forall i \in \{1, 2\}$$

which define a non-compact set. The utility function [71, Example 5.9 (i)] is

$$u_1(x_1, x_2) = \frac{-\sum_{i=1}^{3}(x_{1i}^2 - x_{1i} + x_{2i} - x_{2i}^2)}{x_{11}^2 + x_{21}^2 + 1}$$

The game has a pure Nash equilibrium:

$$x_1^\star \approx (0.8914, 1.1219, 0.8914),$$
$$x_2^\star \approx (0.8914, 1.1219, 0.8914).$$

Example 77. *A general-sum 2-player game in* 3×3 *variables subject to the constraints*

$$x_i \in \left\{z \in \mathbb{R}^3 \,\middle|\, z_1 \geq 0, z_1 z_2 \geq 1, z_2 z_3 \geq 1\right\} \quad \forall i \in \{1, 2\}$$

which define a non-compact set. The utility function [71, Example 5.9 (ii)] is

$$u_1(x_1, x_2) = \frac{-x_{11}^4 - x_{12}^4 + x_{21}^4 + x_{22}^4 - x_{11}x_{13} - x_{21}x_{23}}{x_{13}^2 + x_{23}^2 + 1}$$

The game has a pure Nash equilibrium:

$$x_1^\star \approx (0.9230, 1.0834, 2.8239),$$
$$x_2^\star \approx (1.0459, 0.9561, 1.3804).$$